Video Test and Measurement

Lecture Outline

- Introduction (40 mins)
  - Why do we test? Tolerances
  - Picture quality assessment
  - Types of testing & Monitoring
  - Accuracy and calibration
  - Standards and specifications
  - Broadcast chain & measurement domains
  - Test equipment
- Signal Domain Tests (50 mins)
  - What can go wrong?
  - Analogue Video
  - Digital Video
- Video Domain Tests (50 mins)
  - What can go wrong?
  - Single channel tests
  - Component tests
  - Test waveforms and displays
  - Noise measurements
- Picture Monitor Grading & Setup (40 mins)
  - Revision of Monitor Principles
  - Monitor Grading
  - Monitor Setup – Adjustments and test signals

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Why Do We Test? - Tolerances

- Why do we test video systems?
  - To ensure that the picture quality for the viewer is acceptable.
- What factors affect picture quality?
  - Quality of the source pictures.
  - Signal distortion/picture degradation introduced by the components of the broadcast chain.
  - Quality of the viewer's television.
  - Viewing conditions.
- Tolerance
  - The amount of distortion which is acceptable.
    - For example: We might say that tolerance for HF loss/gain is 10%. This means that the amount of HF loss/gain should be 10% or less.
  - Tolerances can be set for the whole broadcast chain.
  - Each component in the broadcast chain can be given tolerance, such that when the distortions caused by each component add up, the total is within tolerance for the whole chain.

Tolerance

- Suppose that subjective tests show that viewers will accept 11% of some kind of impairment.
- The total tolerance could be allocated to a notional broadcast chain, as follows:

Camera	Amplifier	Record / Replay	Transmitter	TV	
1%	0.1%	1%	7%	1%	**Total: 10.1%**

- If each part of the chain is measured to ensure that it is within tolerance, then the total impairment for the whole chain should ensure that the effect of the impairment is of an acceptable level.
- **Conclusion:** You may well not see unacceptable levels of impairment if you are only looking at part of the broadcast chain.

Picture Quality Assessment - 1

- The quality of a TV picture can be assessed by people looking at picture monitors / TVs.
  - Experts
  - Members of the public (Non-expert)

- It is helpful to have a standard way of describing visual picture distortion.
  - EBU created a grading scale – now described in ITU-R BT.500 (Methodology for the subjective assessment of the quality of television pictures)

Five Grade Scale			
Quality		Impairment	
5	Excellent	5	Imperceptible
4	Good	4	Perceptible, but not annoying
3	Fair	3	Slightly annoying
2	Poor	2	Annoying
1	Bad	1	Very annoying

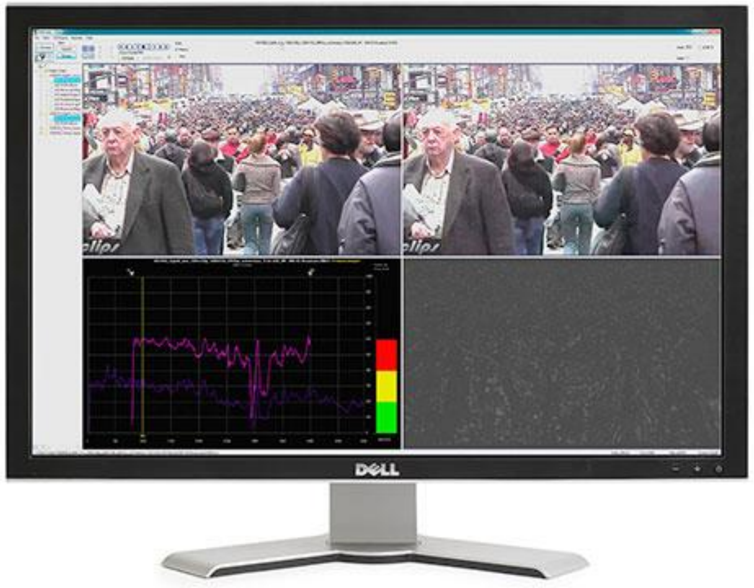
1 – very Annoying

Picture Quality Assessment - 2

- ITU-R BT.500 describes:
  - Type of assessment
    - Under optimum conditions - These are typically are called quality assessments.
    - Under non-optimum conditions – These are typically are called impairment assessments.
  - Assessment methods
    - For example the EBU Double-stimulus Impairment Scale
  - Viewing conditions
    - Laboratory (reference) environment.
    - Home environment.
  - Monitor setup, resolution and contrast (ITU-R BT.814 and BT.815)
  - Selection of test material.
  - Observers and their instructions.
  - Presentations of results.

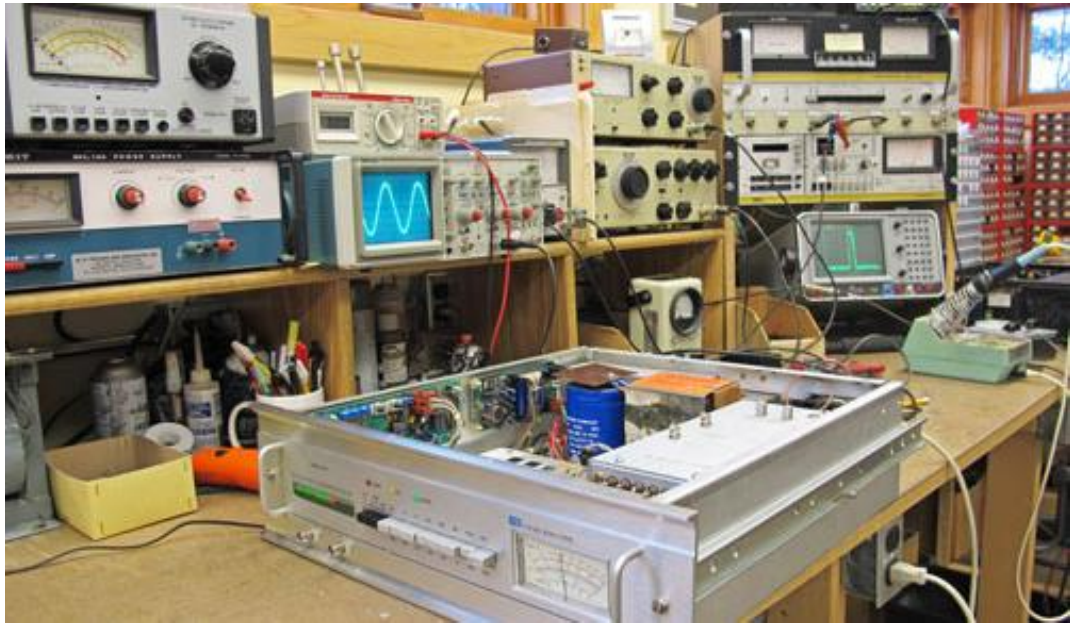
Picture Quality Assessment - 3

- ITU-R BT.500 was designed with CRT displays in mind.
- For flat panel displays it is supplemented by:
  - ITU-R BT.2022
    - General viewing conditions for subjective assessment of quality of SDTV and HDTV television pictures on flat panel displays.
  - ITU-R BT.2129
    - User requirements for a Flat Panel Display (FPD) as a Master monitor in an HDTV programme production environment.
- Automatic Picture Quality Assessment can be done for certain types of impairment such as noise and coding artefacts.
  - For example: Tektronix PQA600C

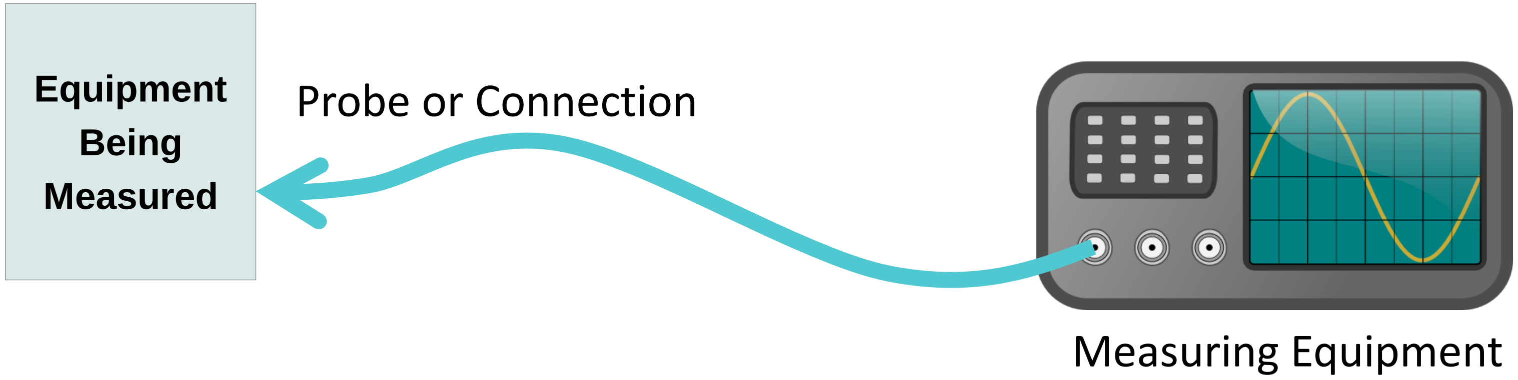


Types of Testing

- Operational monitoring and testing
  - Monitoring ‘live’ signals in a studio, OB or playout area.
  - Quality checking final programmes for delivery to playout.
- System Testing
  - Offline testing of a piece of equipment or system as part of fault finding or routine maintenance.
- Acceptance Testing
  - Testing of a piece of equipment or system before it can be accepted for something, such as sale, purchase or before it is used for broadcasting.



Accuracy and Calibration

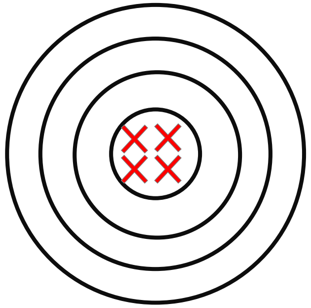


- How accurate is the reading? What are the possible sources of error?

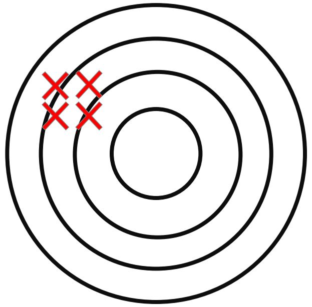
  1) Measuring Equipment.
     - Precision – how many decimal places?
     - Accuracy – But is it right?
     - Calibration – Equipment is compared to a reference and setup up to be 'correct'

Accuracy = Precision + Calibration

  2) Loss/gain and distortion in the probe/cable.
  3) Loading – connecting the probe changes the level.



Accurate



Precise (but not accurate)

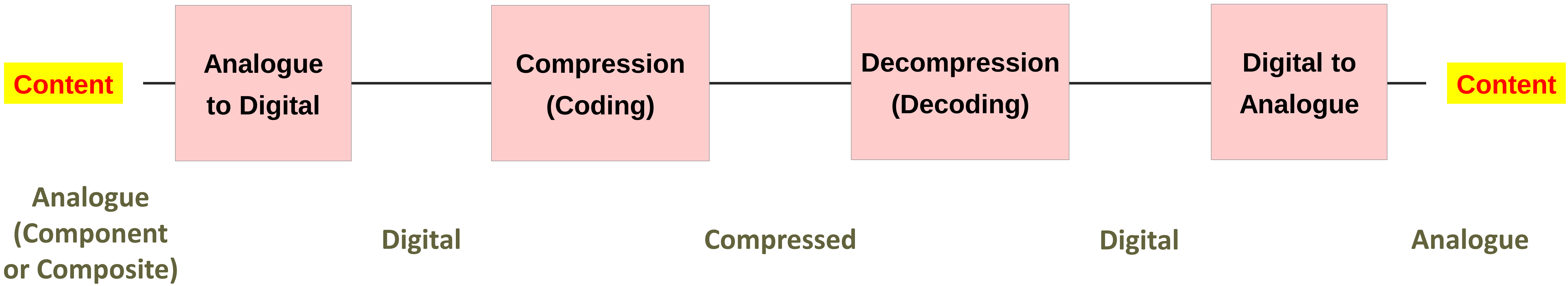
Standards and Specifications

- When testing, we are usually comparing the results to a reference, such as a standard or specification.

- Video Signals and Formats
  - Video standards such as ITU-R BT.709
  - File format standards such as MXF
  - Stream/data formats such as MPEG Transport Streams
- Video Equipment and Systems
  - Manufacture specifications
  - Broadcasters’ acceptance criteria
  - Installation project requirements/specifications
- TV Programmes
  - Programme delivery specification such as AS-11 DPP

- These can overlap, and one standard may refer to another.

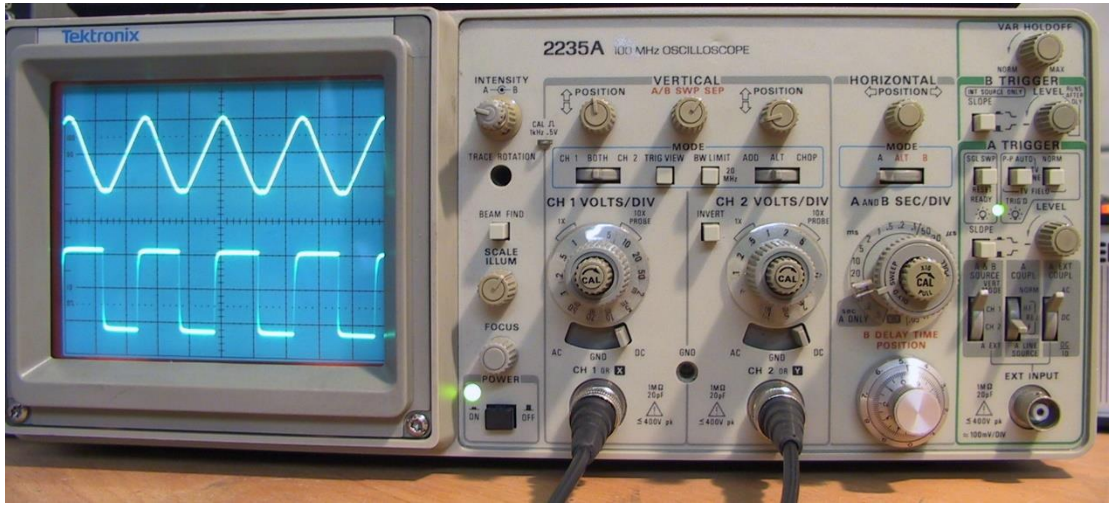
Signal Stages

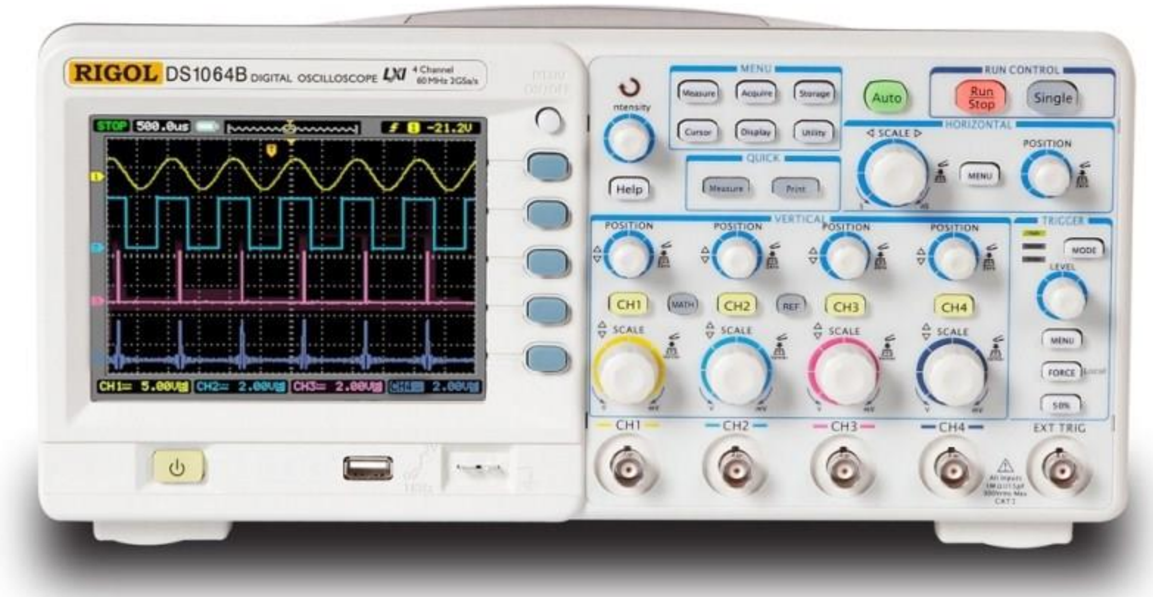


- Each stage has appropriate tests and measurements that can be made.
- For example:
  - Analogue: HF loss, Chrominance-Luminance Delay, Video Level, Gamut
  - Digital: Eye Height, Data Format
  - Compressed: Coding Artefacts
  - Content: Photo Sensitive Epilepsy (PSE)
- The above diagram is greatly simplified.

Test Equipment - Oscilloscopes

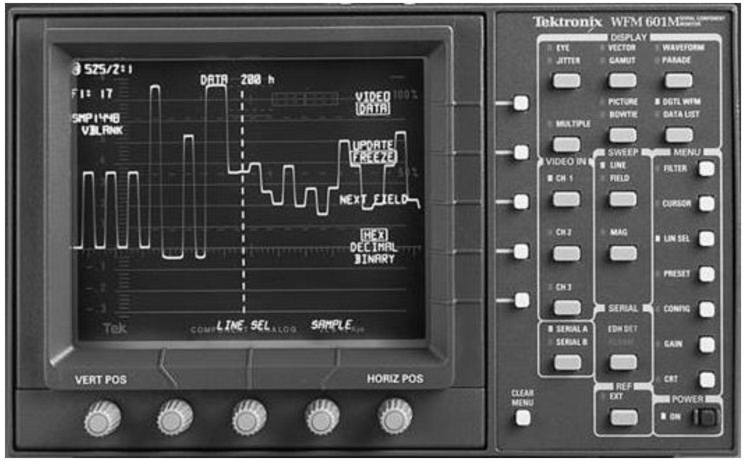
- Analogue or digital processing.
- CRT or LCD display.
- Bandwidth for Analogue TV: >10MHz (SD) >40MHz (HD).
- Often have TV trigger modes.
- High bandwidth scopes may allow eye display for SDI.
- Input impedance usually >1MΩ. Need to use T-piece and 75Ω termination for video.

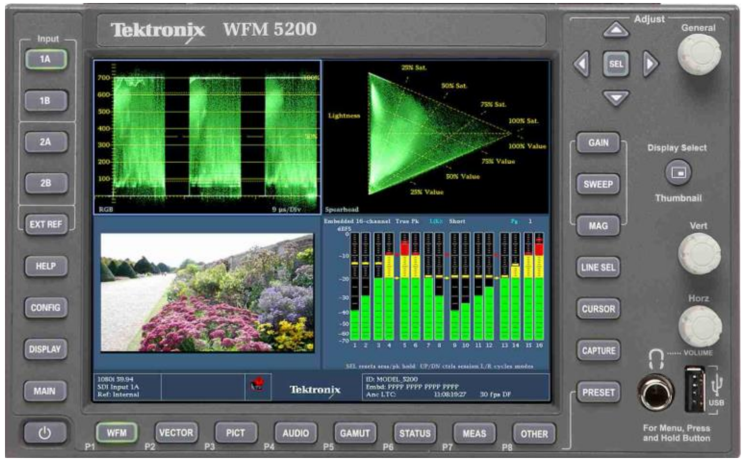




Test Equipment – Waveform Monitors

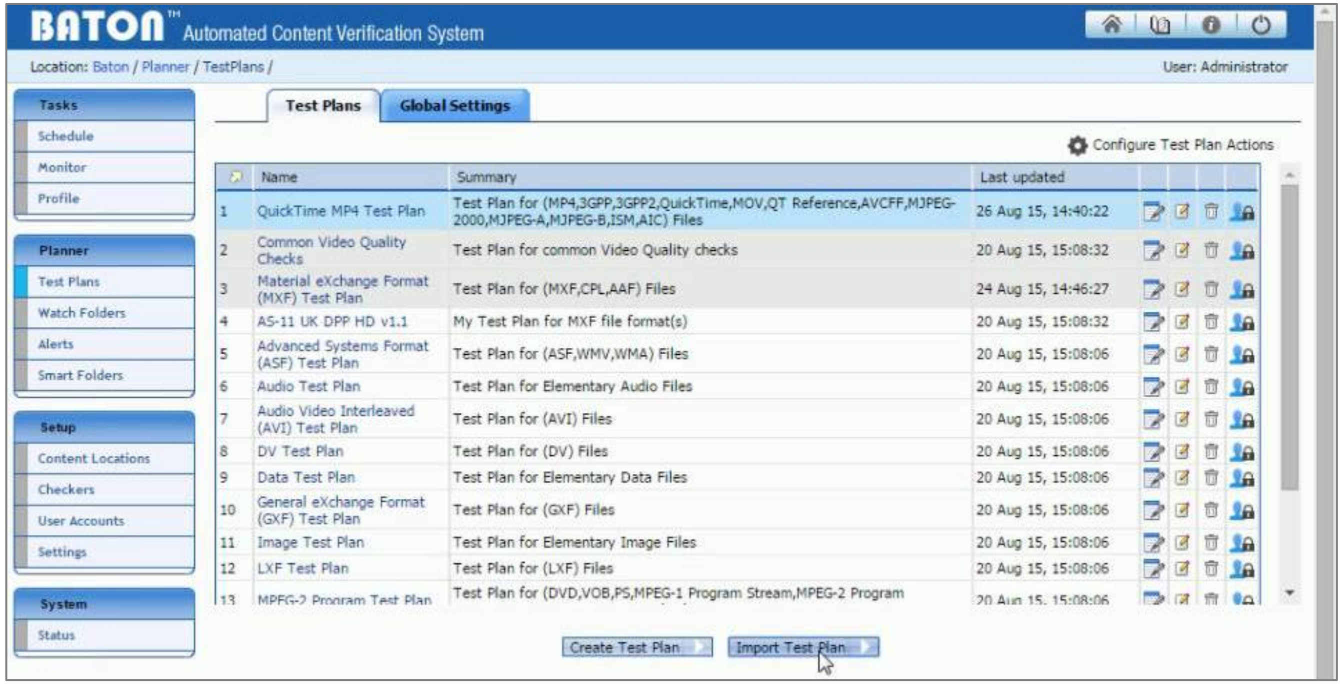
- Available for analogue composite, analogue component and SDI video signals.
- Older designs had CRT displays. New designs tend to have LCD type displays, or connect to external computer monitor.
- May also be implemented in software for use with a computer with a video input, or with editing software.
- 75Ω inputs – active or passive loop-through.
- Offer various display/measurement options:
  - Waveform, Parade, Vector, Lightning, Bowtie, Arrow/Gamut, Eye Display, Picture, Signal Data, Ancillary Data, Embedded Audio.



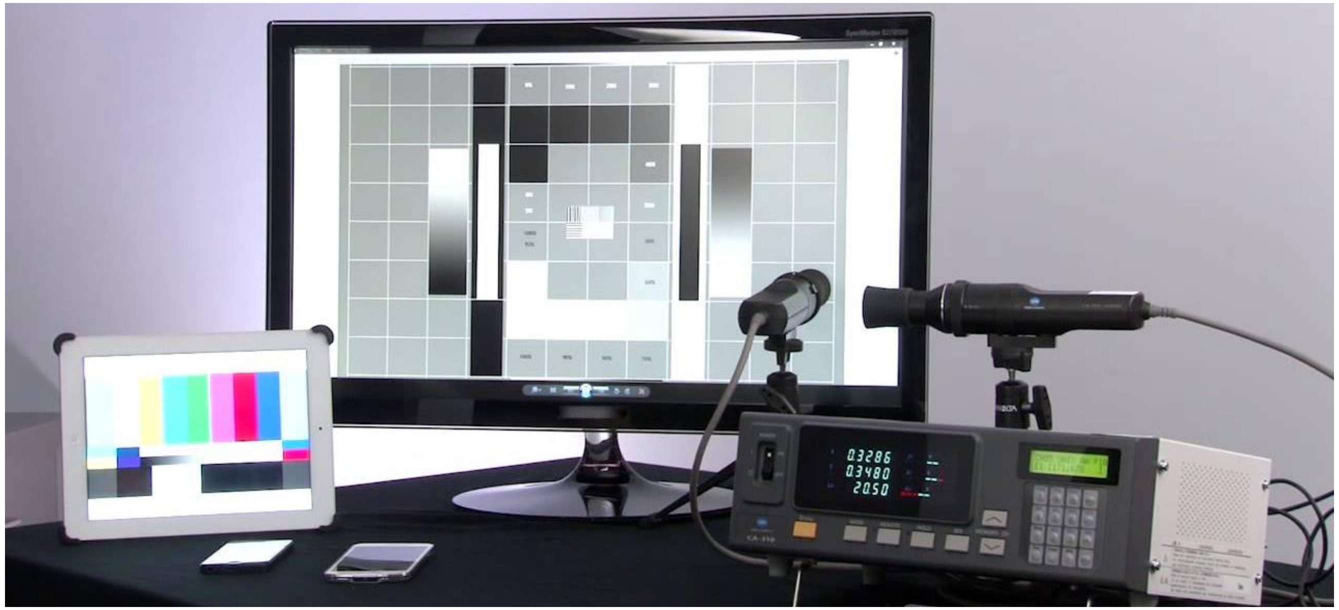


Test Equipment – AQC and Monitor Testers

- Automatic Quality Checking
  - Automatically check media files before broadcast. Can check signal format, metadata, content.
  - Examples:
    - Interra Systems – Baton,
    - Tektronix – Cerify.



- TV Monitor Analyzer
  - Some monitors have this feature built in (require an external probe).
  - Good for accurate, repeatable setup of displays.



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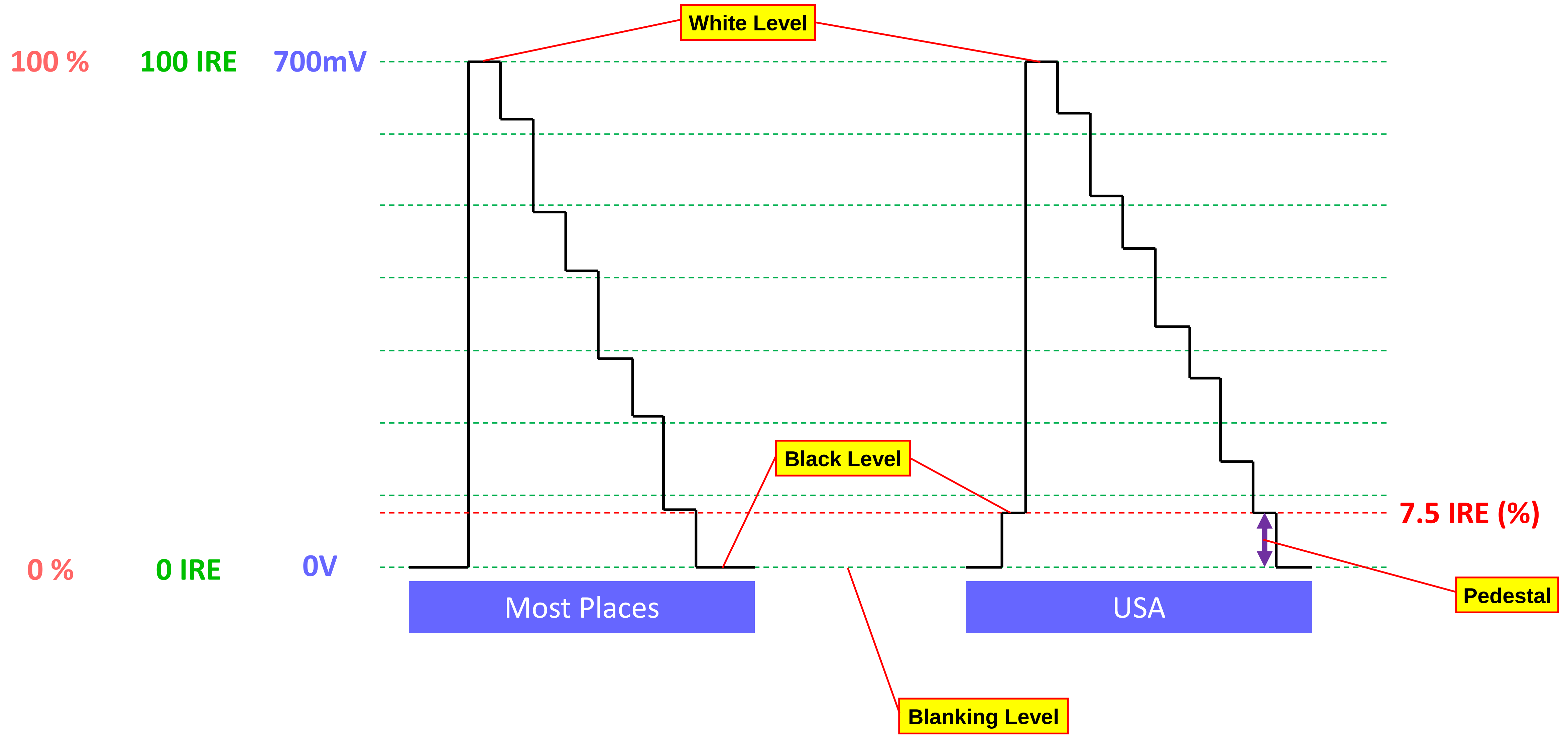
Signal Domain - What Can Go Wrong…

- Signals are distributed along wires, fibres or radio frequency “wireless” links such as WiFi, terrestrial transmitters or satellite links.
- We will call the link a channel.
- The signals carried by the channel may carry **analogue** information or **digital** information.

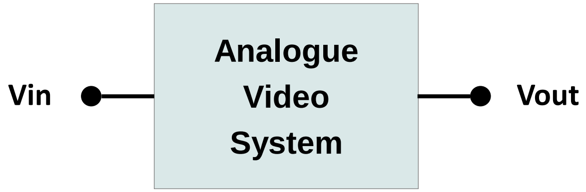
- To accurately pass the signals, we require the following of the channel:
  - **Gain** and **propagation delay** should be **independent of frequency**.
    - If this is not the case it is called **Linear Distortion**.
  - **Gain** and **propagation delay** should be the **same for all input levels**.
    - If this is not the case it is called **Non-Linear Distortion**.
  - **Unwanted signals** will **not be added** by the transmission channel
    - Unwanted signals are called **Noise**.

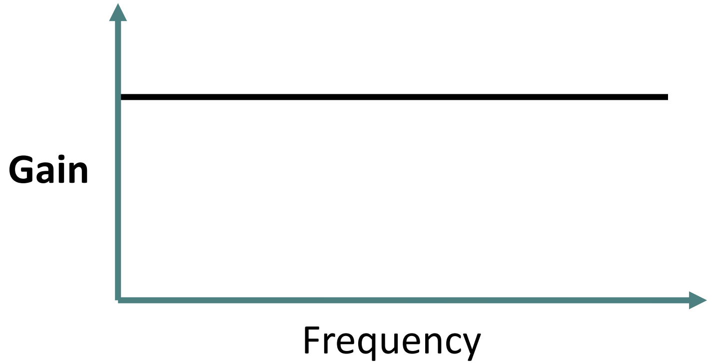
Analogue Video Levels

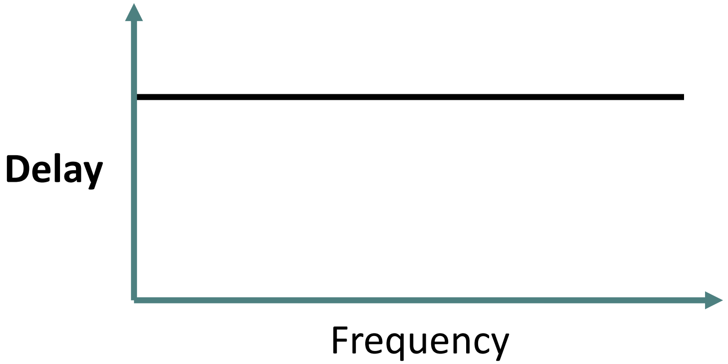


IRE = Institute of Radio Engineers

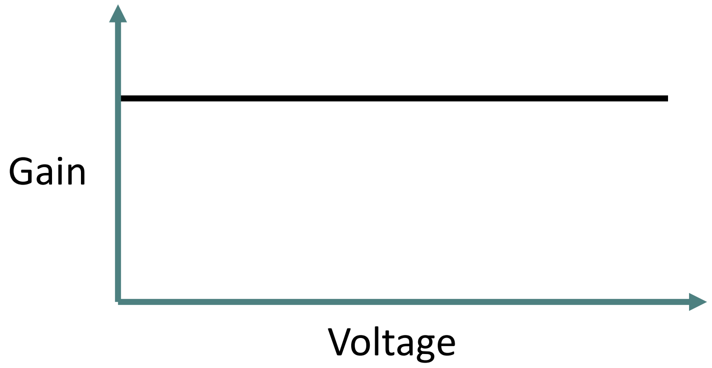
Signal Domain – Analogue Video - Linear and Non-Linear Distortions

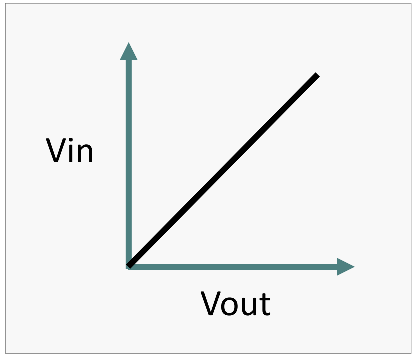


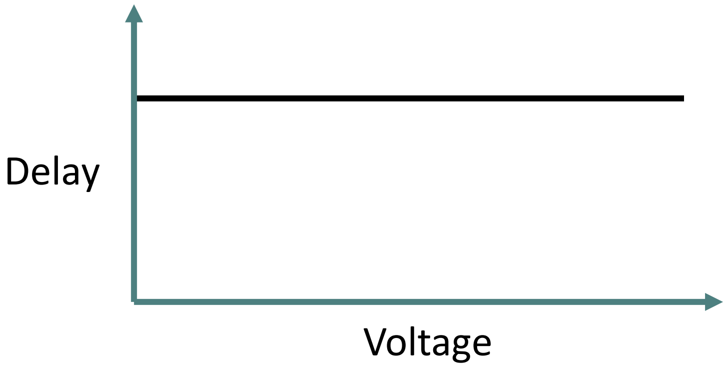




Response of a Channel with No Linear Distortions

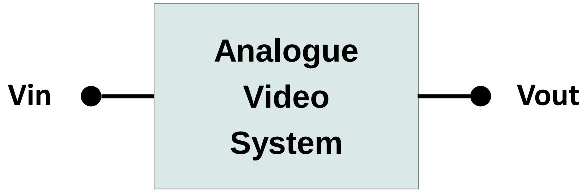


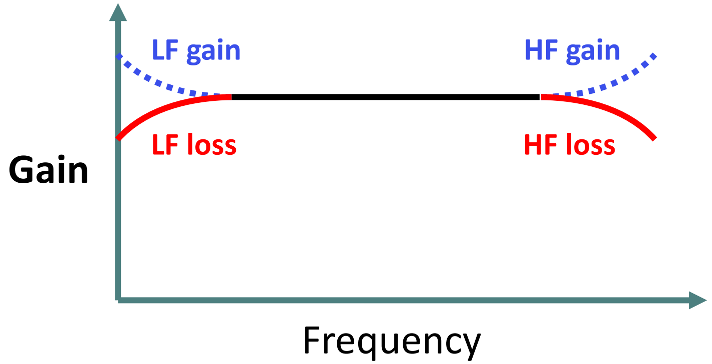


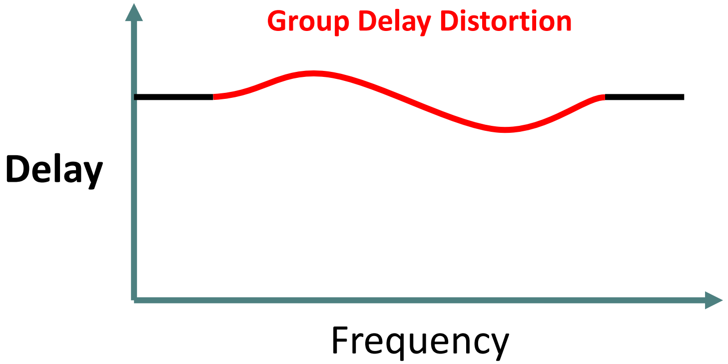


Response of a Channel with No Non-Linear Distortions

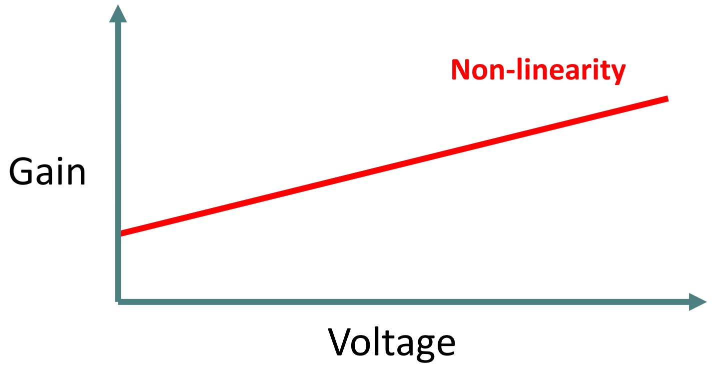
Signal Domain – Analogue Video - Linear and Non-Linear Distortions

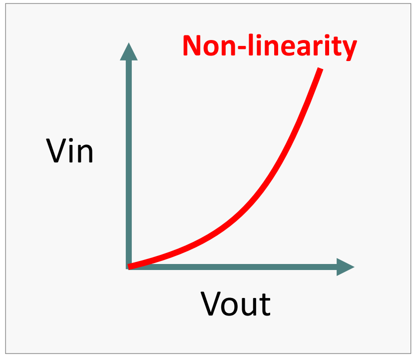


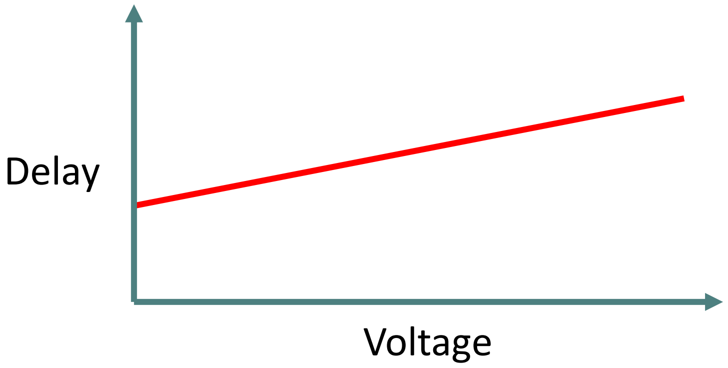




Linear Distortions



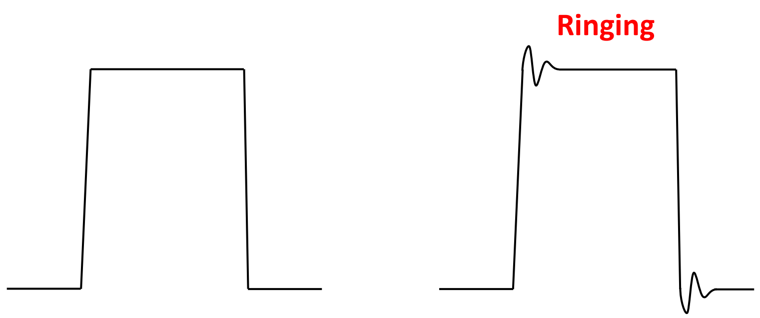




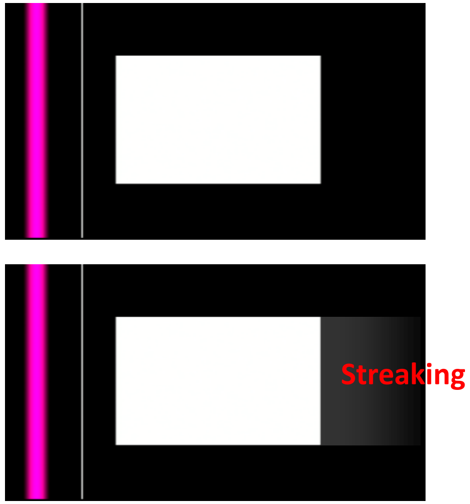
Non-Linear Distortions

Analogue Video - Distortions

- When the channel is carrying analogue video, distortions will directly affect the picture (video signal domain).

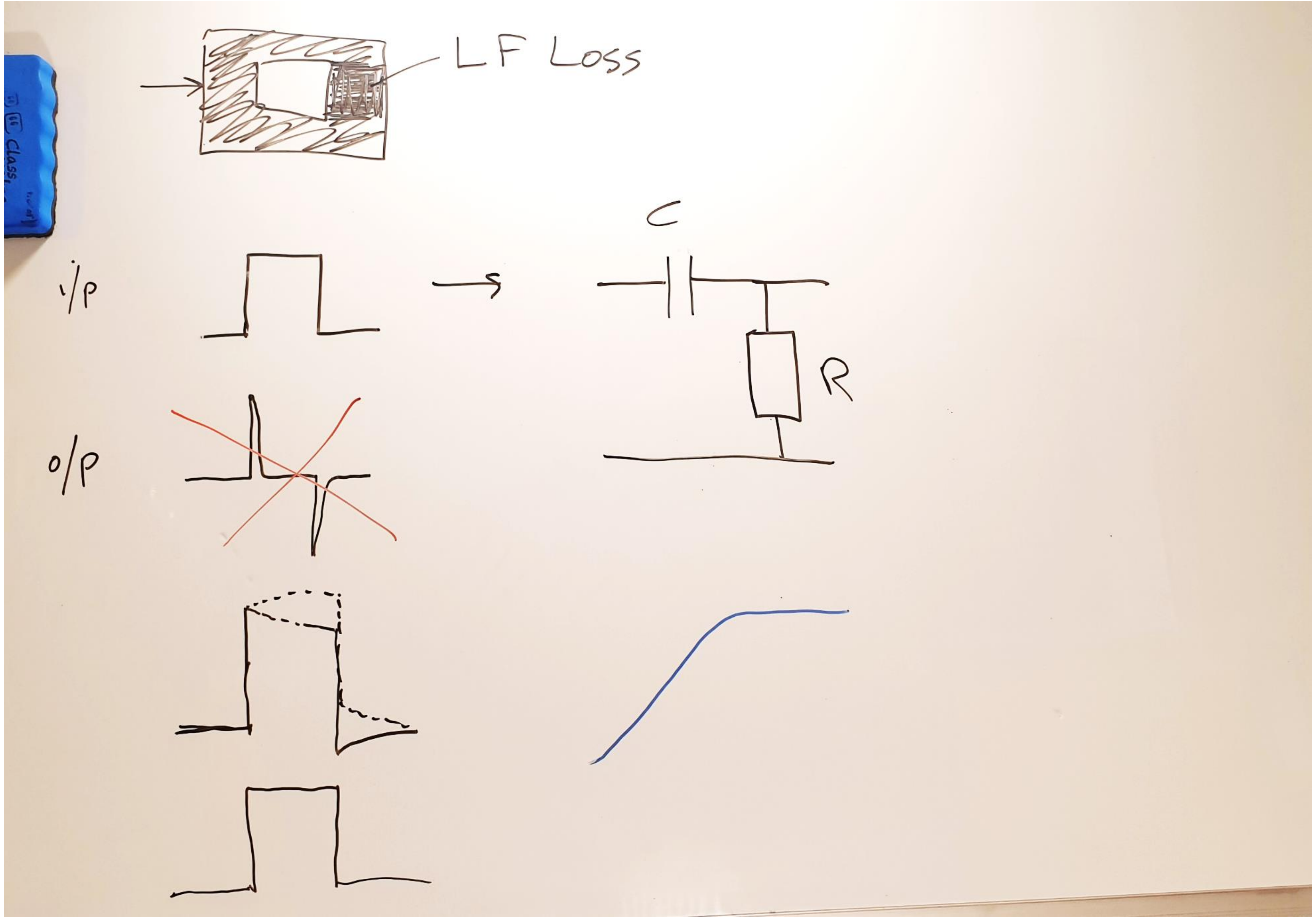


- HF Gain/Loss
  - **Over-sharp** / **soft** pictures. Possibility of **ringing** or **overshoots**.
  - Chrominance/Luminance gain errors in composite signals.
- Group Delay Errors
  - **Different frequencies** are **delayed** by **different** amounts.
  - Effect is **ringing/smearing** on edges (which can look like soft pictures).
- LF Gain/Loss
  - Light/dark **streaking** after bright objects.



- Non-Linearity
  - Black or white **crushing**/clipping.
  - Differential Gain/Phase in composite systems (Chrominance gain/phase varies with Luminance level).

Cause of LF Loss



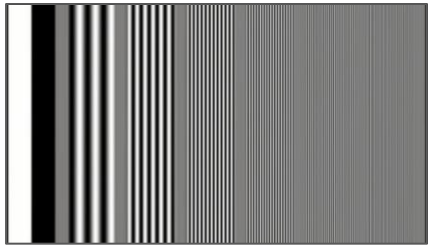
Analogue Video Test Signals

- With analogue video, there is a direct correlation between the Signal Domain and the Video Domain. Therefore, most of the tests cover both.

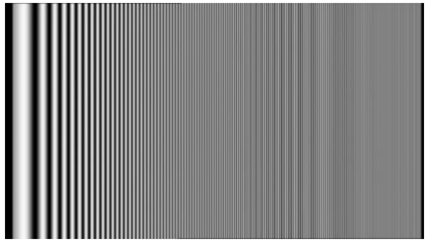
- Frequency Response Tests
  - Multiburst
  - Line Sweep
  - Zone Plate

- Frequency Response Tests
  - Pulse & Bar – Designed be representative of real pictures.
  - 50Hz Square Wave – For testing low frequency response.

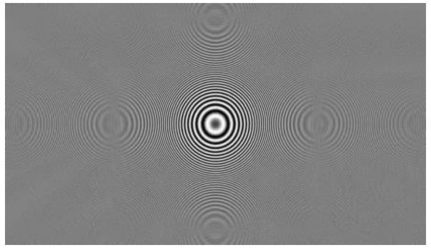
- Non Linearity Tests
  - Ramp/Sawtooth
  - Staircase
  - Composite Ramp/Staircase



Multiburst



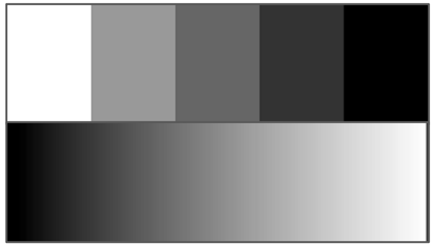
Line Sweep



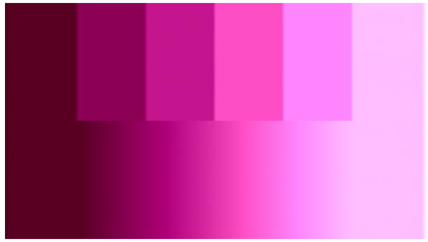
Zone Plate

Pulse & Bar

50Hz Square Wave



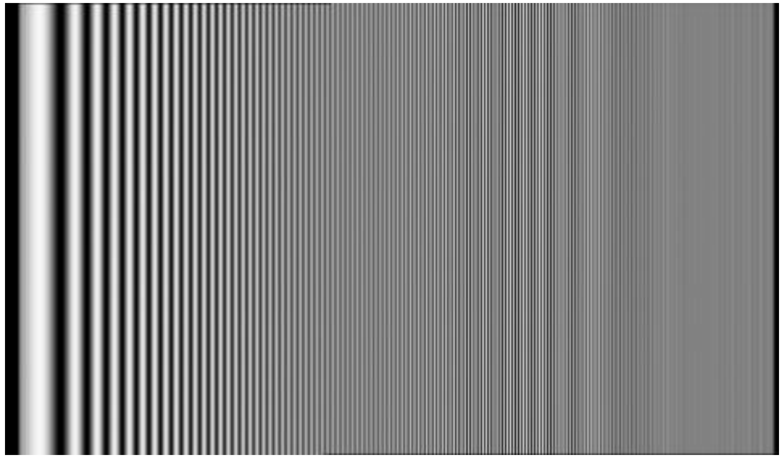
Staircase & Ramp



Composite Staircase & Ramp

Pulse and Bar

- To measure frequency response, you could use Line Frequency Sweep.



Line Sweep

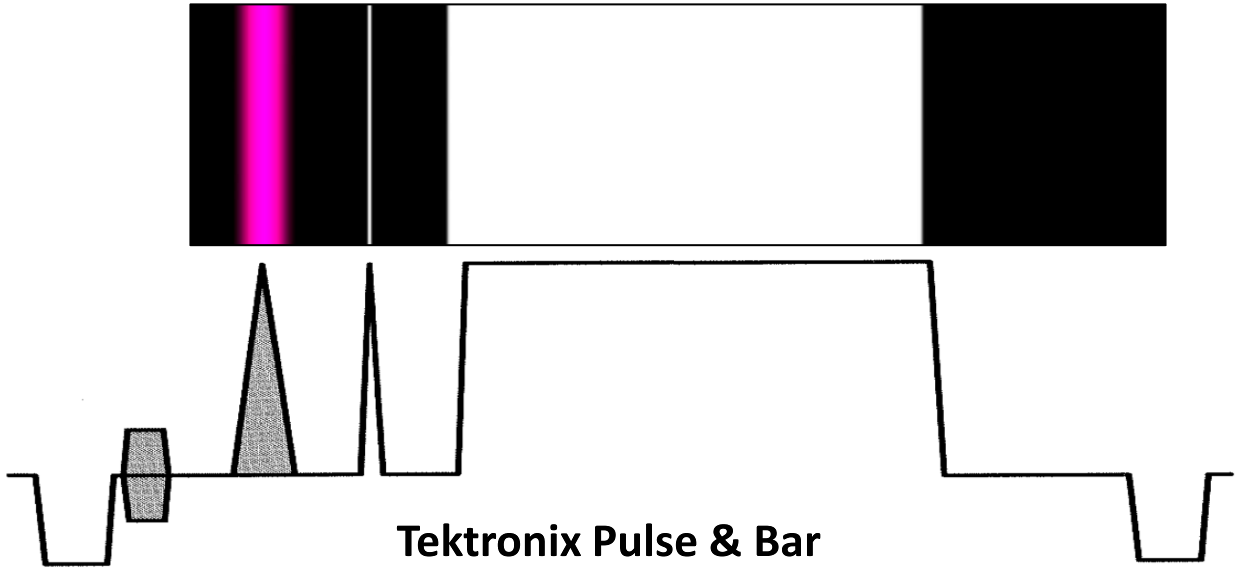
Pulse and Bar

- **Pulse and Bar** is designed to measure frequency response in a way that gives a **better indication** of the effect of distortion on **real pictures**.
- It has **3** main parts:

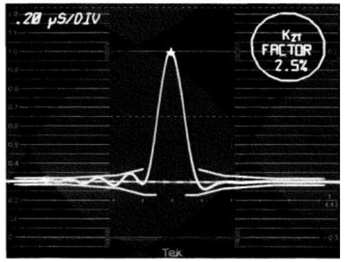
1) **2T Pulse**
   - Good for showing HF Loss/Gain and Group Delay Distortion
     - Pulse amplitude (pulse to bar ratio) – HF
     - Pulse ringing – Group Delay

2) **Bar**
   - Bar Tilt – LF Loss/Gain

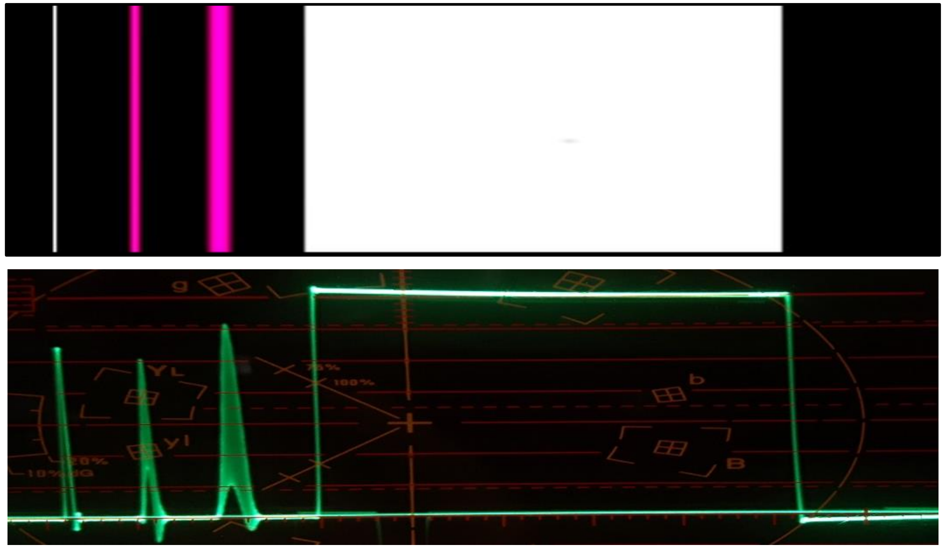
3) **Composite Pulse**
   - Added to show the effects on the Chrominance



Tektronix Pulse & Bar

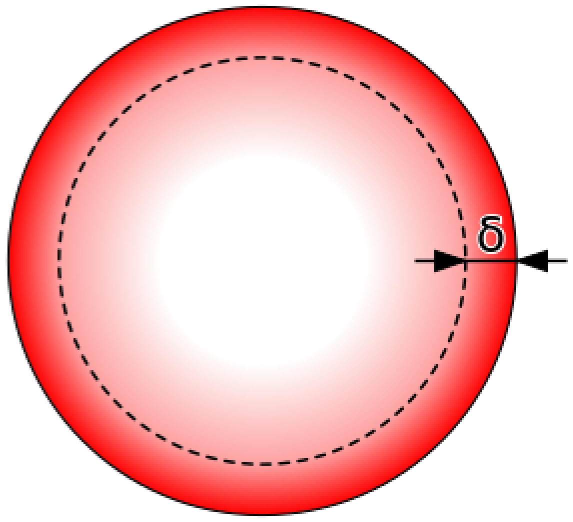


2T Pulse Showing Effects of Group Delay



Distorted BBC Pulse & Bar

Common Causes of Analogue Distortions



- A common cause of HF Loss is long cable runs.
  - Skin Effect causes the resistance of the cable to increase with frequency.
  - Current flows only along the outer part of the cable (the skin).
  - Cross-sectional area of the skin reduces with frequency.
  - Can be compensated for with equalisation
  - Several km of transmission is possible, if properly equalised

- A common cause of LF loss/gain is dried up capacitors in old analogue equipment

Signal Domain Tests - Digital Video (SDI)

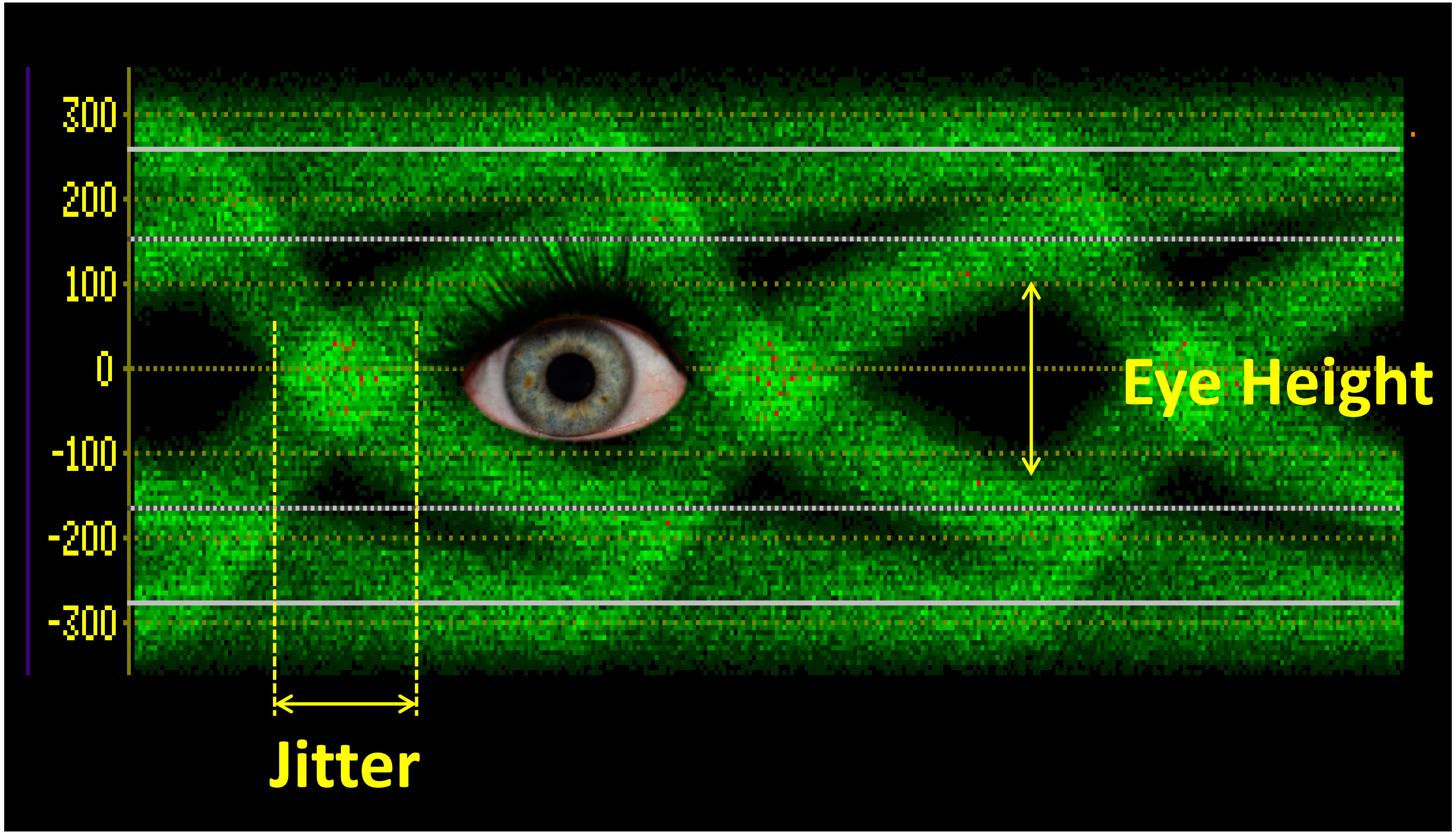
- Small distortions of the signal channel are less important.
- Nearly all SDI equipment is auto-equalising.
- Distances of up to 200m for HD, 600m for SD are typically possible.
  - Very dependent on the cable type.
- SDI is very prone to errors caused by signal reflections from unterminated/damaged cables.

- An SDI receiver needs to be able to:
  - distinguish between a high and a low voltage.
  - Determine where the changes take place.

- Eye display and the Eye Height allow us to assess the likelihood of this being the case.

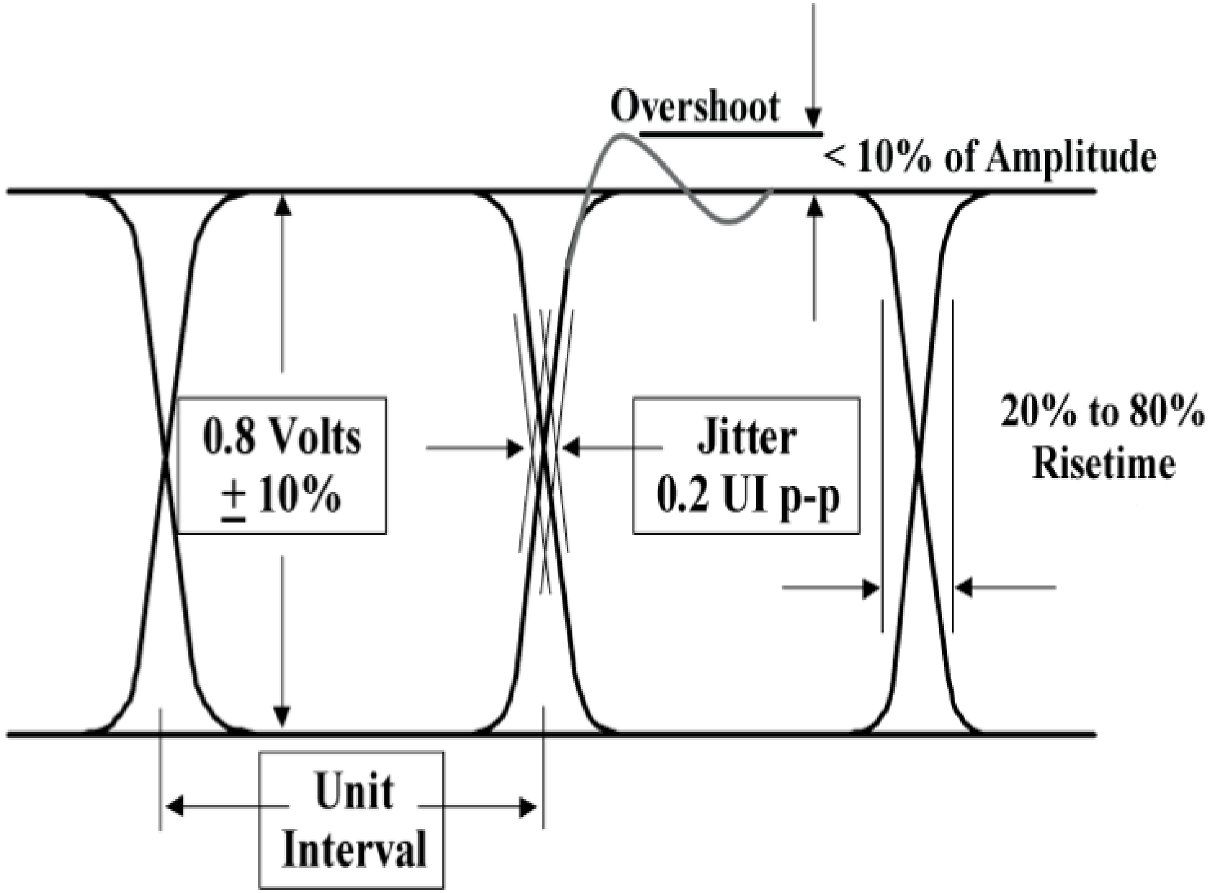
Eye Display, Eye Height and Jitter

- Eye display shows the voltage on the wire – triggered on the clock edges.
- Eye height represents signal integrity.
- Display also gives an indication of Jitter (varying delay).



- Some waveform monitors will show graphs of jitter, with various filters applied.

SDI Eye-Height



SDI Pathological Testing

- A picture designed to make it likely that, after scrambling, there will be a pattern of digits which is most challenging to the receiver.

- There are 2 common types:
  - Cable Equalizer Test
    - Y = 198h C = 300h (a Magenta)
    - The repetition of 19 zeros followed by 1 one or 19 ones followed by 1 zero will test the ability of the equalizers, input buffers and coupling network (AC-coupling capacitor or transformer) to handle baseline wander and low frequency content.
  - Phase Locked Loop Test
    - Y = 110h C = 200h (Dark Grey)
    - The repetition of 20 zeros followed by 20 ones will test the ability of the Receive Clock and Data Recovery (CDR) PLL to remain locked for patterns with very low transition density. Moreover, the transition from high transition density to low transition density affects the gain and bandwidth of the PLL, making it a challenging pattern to recover with no errors.

- They can be combined in a single test pattern, often called SDI Checkfield.

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Video Domain Tests – What Can Go Wrong…

- We have **already** looked at **video domain** test for **Analogue video**.
  - because the **video domain** and **signal domain** for analogue are essentially the **same**.
- The **video domain** of an **SDI video signal** can potentially suffer from the **same problems as analogue** video
  - Although it's usually **much less likely**.
- So we can use the **same techniques** to look at the frequency response, gain, delay response and noise (including **quantisation noise**) of **each component** channel within the SDI signal.

Signal Domain Tests - Component Video

- With any component video (carried as SDI or analogue), there can be **gain/delay errors** between the 3 components.

- **RGB** is **much less robust** in this respect than **Y, Pr, Pb** because such errors cause errors in the luminance signal – human vision is most sensitive to this.

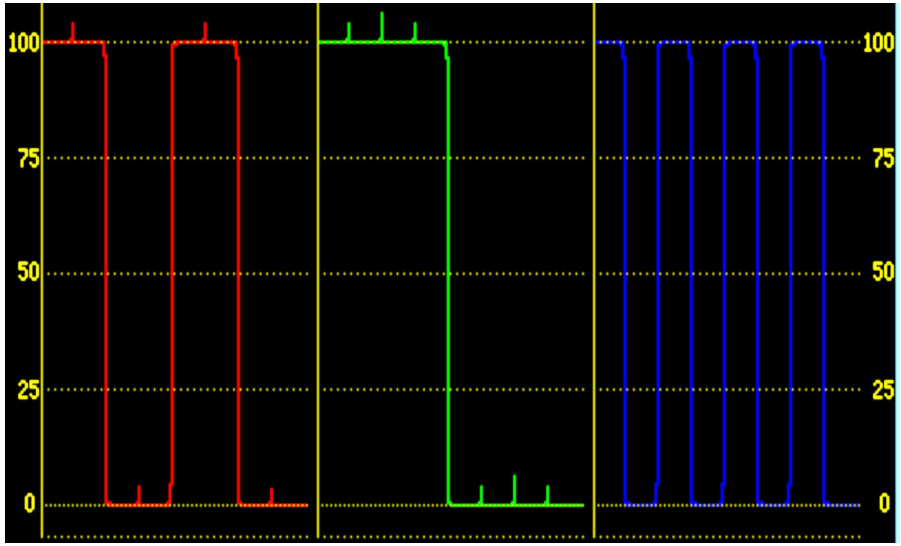
- Possible causes of component video inter-channel gain and delay errors include:
  - Incorrect cable lengths / terminations
  - Incorrect adjustment of timing within equipment using digital processing

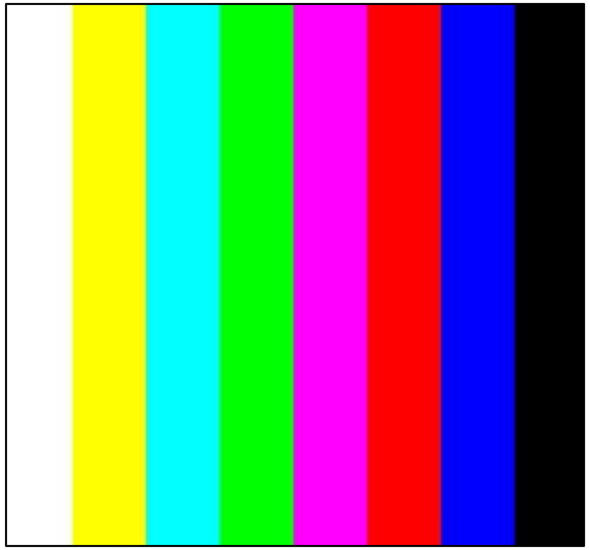
Signal Domain Tests - Component Video

- Test Waveforms:
  - Bowtie (requires a special measurement mode on the waveform monitor)
  - Colour Bars (100% and 75%).
- Display Modes:
  - Vector
  - Lightning
  - Parade/Overlay

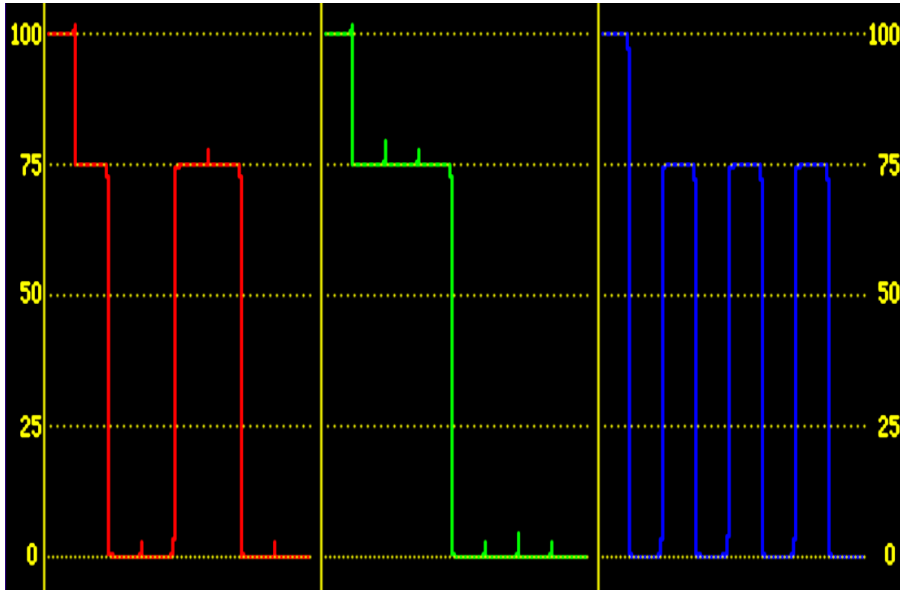
Test Waveforms - Colour Bars

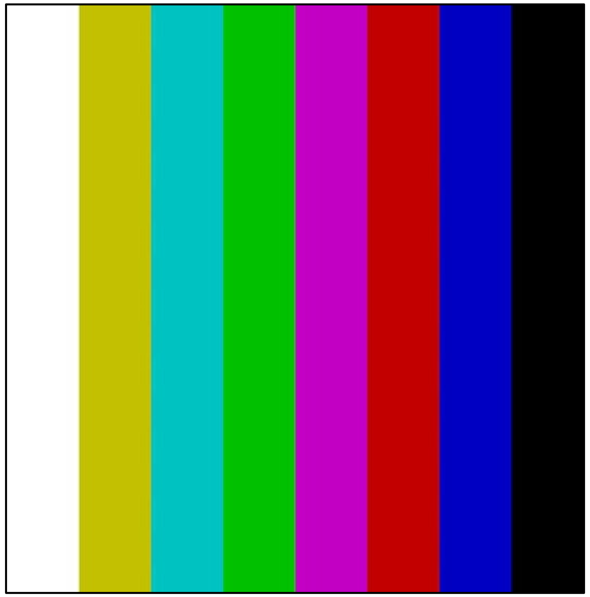
- **100% colour bars** have the primary and secondary colours, black and white.
- The coloured parts are **100% saturated** and **100% amplitude**.



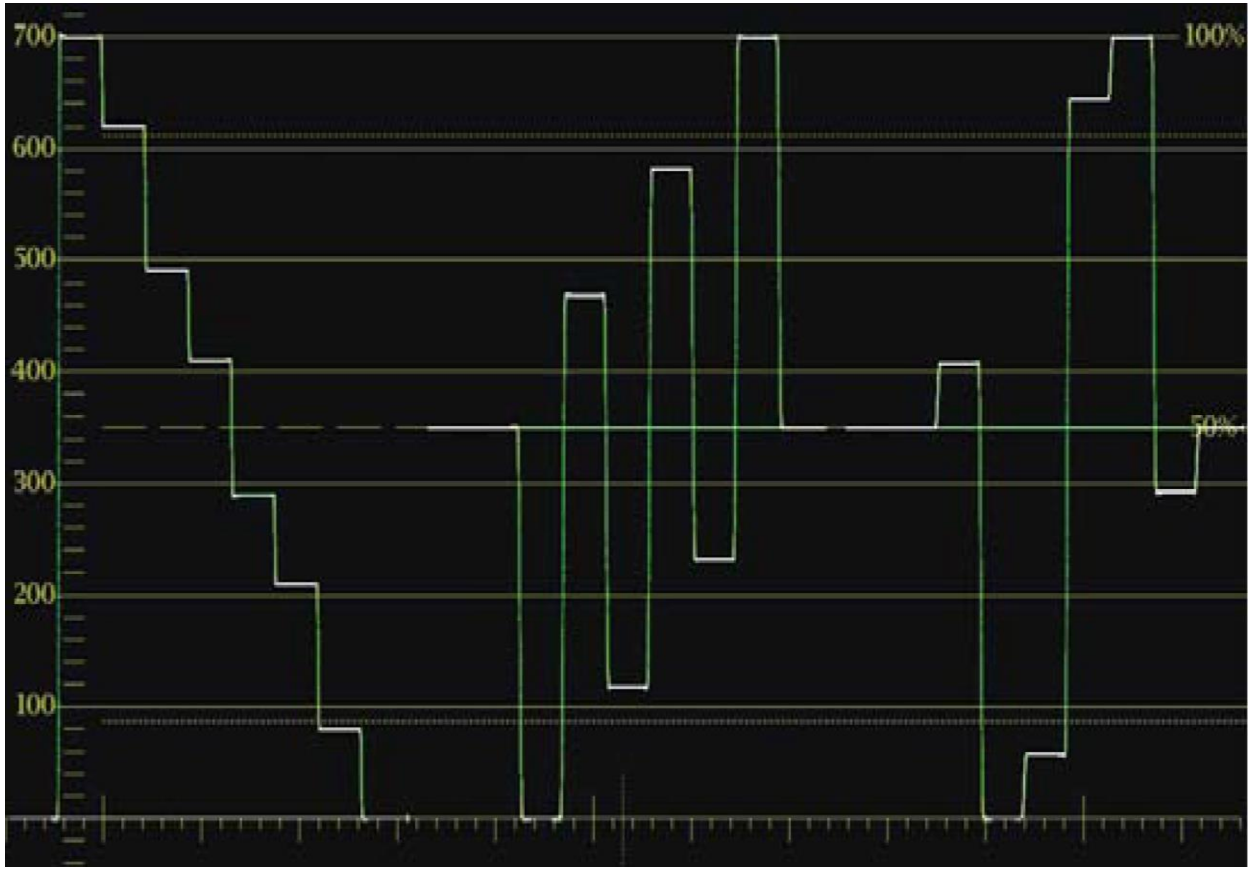


- **75% colour bars** (EBU Bars) have the same colours as 100% bars.
- The coloured parts are **100% saturated** but only **75% amplitude** (they are **darker** than the colours in 100% bars).





Test Waveforms - Colour Bars

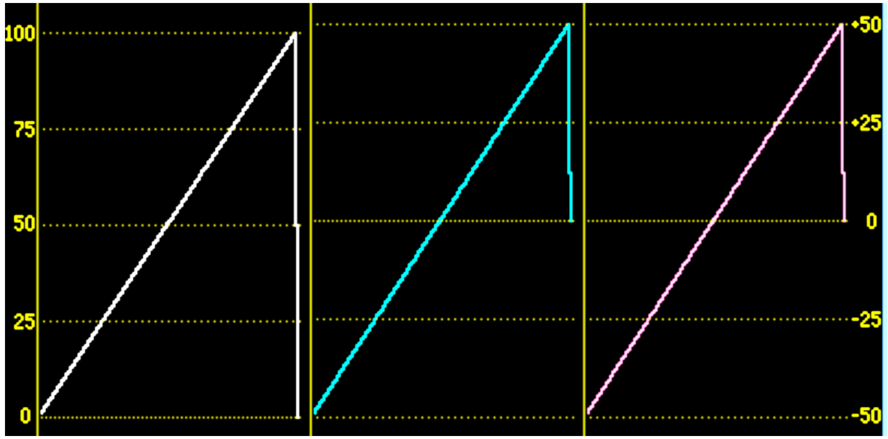


100% Colour Bars (Rec 601)

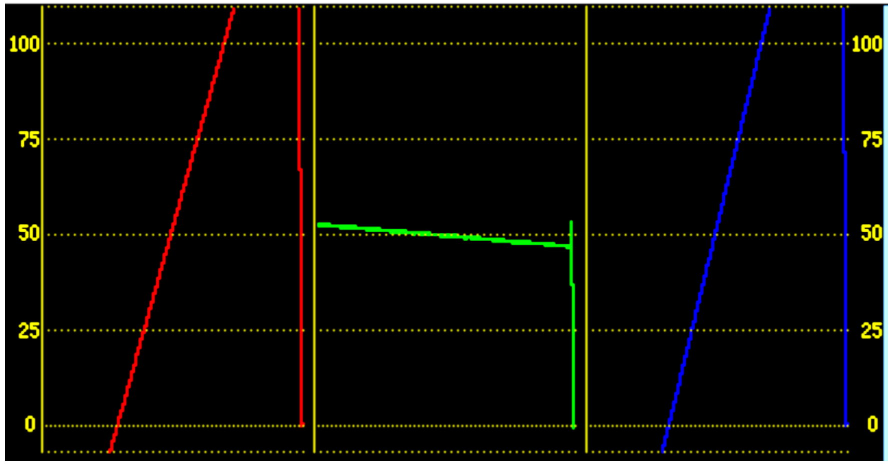
100% Colour Bars – HD (Rec 709)

Gamut and Valid Signals

- The full range of reproducible colours in a TV system can be represented by combinations of Red Green and Blue signals between 0% and 100%.
- It is possible to create Y, Pr, Pb component or PAL/NTSC signals within their specific limits, yet converted back to RGB they result in signals above 100% or below 0%.
- Another cause of such signals could be video sources, such as cameras, which do not clip the signals at 100%.
- This is called a Gamut error.
- Such signals are called invalid.

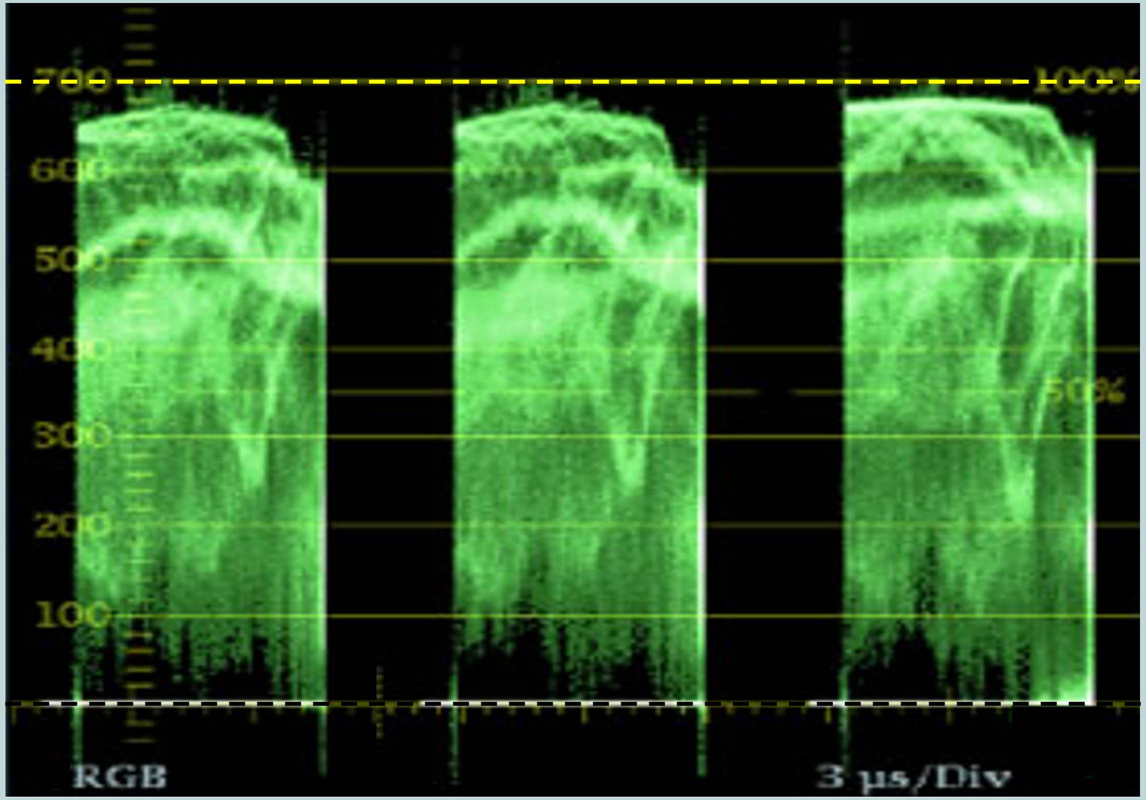


A Ramp signal on all the components (Y, Pb, Pr). All components are within range.

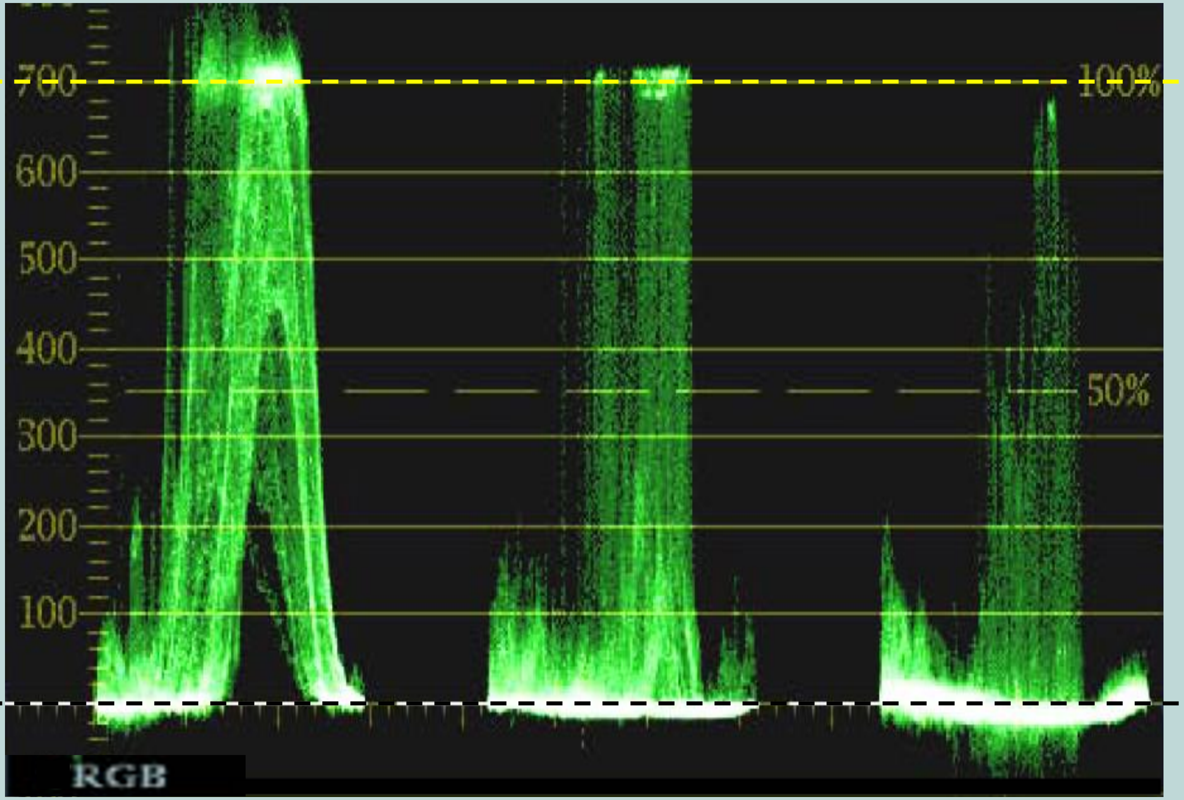


The same signal converted to RGB. R and B are well out of range (Gamut Error)

Gamut Error Shown on an RGB Parade Display

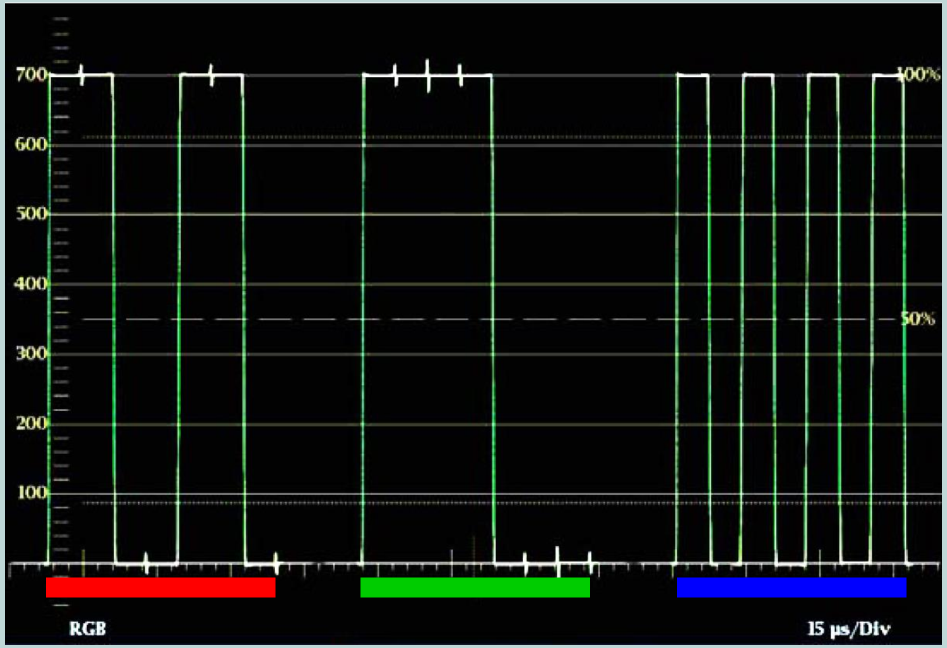


Gamut OK

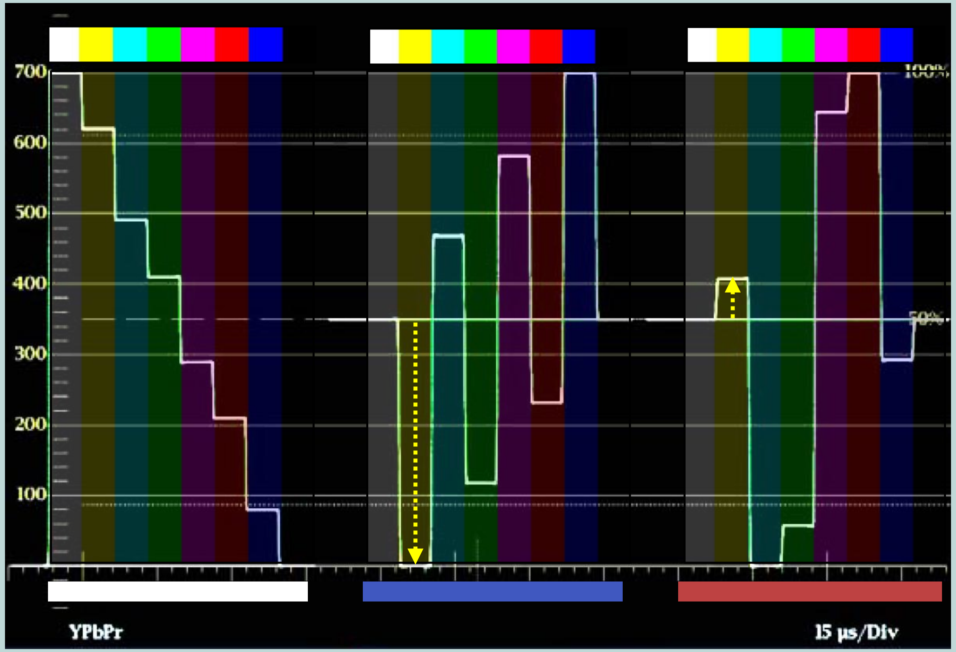


Gamut Error

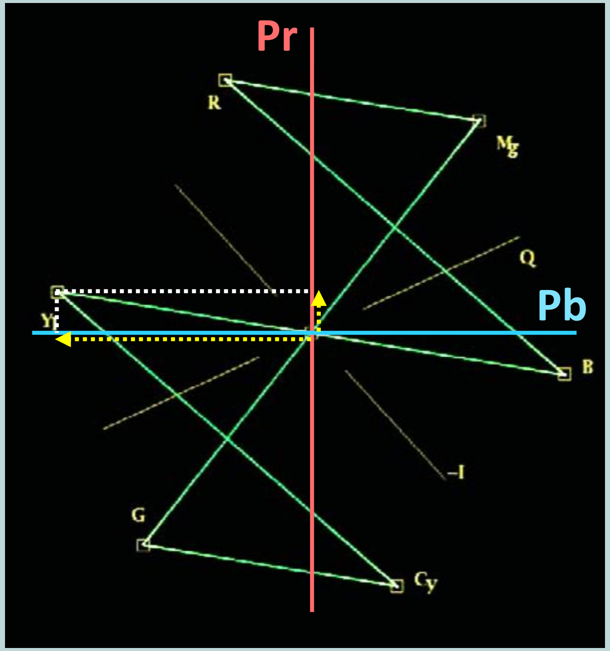
Waveform Monitor Display Modes - 1



RGB Parade Display – 100% Colour Bars



Y, Pr, Pb Parade Display – 100% Colour Bars



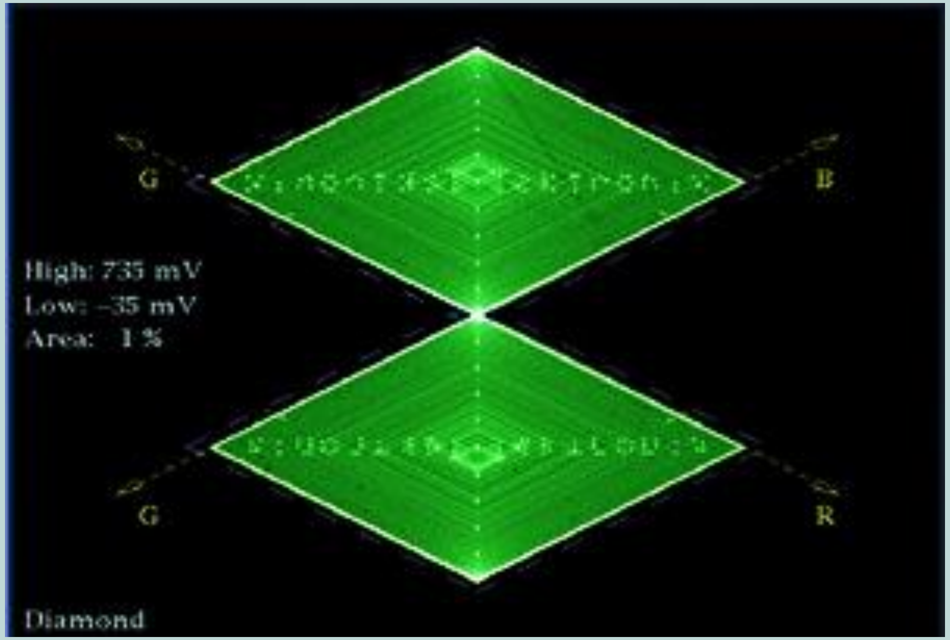
Vector Display – 100% Bars

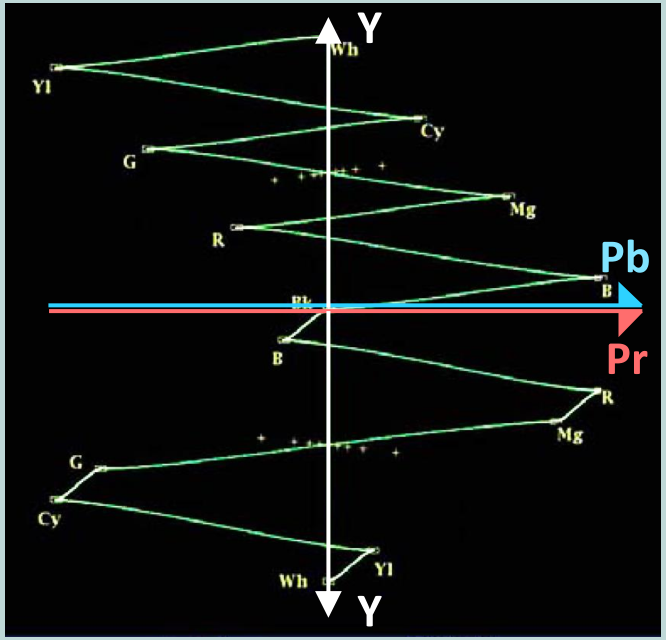
- **1 line** worth of time – **Red**, **Green**, **Blue**.
- Commonly used in editing and post production.

- **1 line** worth of time – **Y**, **Pb**, **Pr**.
- Pr and Pb waveforms have been lifted up to allow comparison with Y
- Useful for general monitoring of component signals (inc. SDI) in TV studios and master control.

- Vertical Axis – **Pr**
- Horizontal Axis – **Pb**
- Y not shown.
- Useful for lining up cameras on grayscale chart.

Waveform Monitor Display Modes - 2





Bowtie Display

- Left side: Y- Pb
- Right side: Y- Pr
- Bowtie waveform consists of:
- 500kHz Y, 502kHz Pr and Pb.
- Designed to cancel out in the centre if levels and delay between components are correct.

Diamond

- Shows RGB levels in a way designed to highlight Gamut errors.
- There is an equivalent display for NTSC and PAL called '**Arrow**'.

Lightning Display – 100% Bars

- Top: Y against Pb
  - Y increasing vertically upwards.
- Bottom: Y Against Pr
  - Y increasing vertically downwards.
- Bendy transitions show delay between channels.

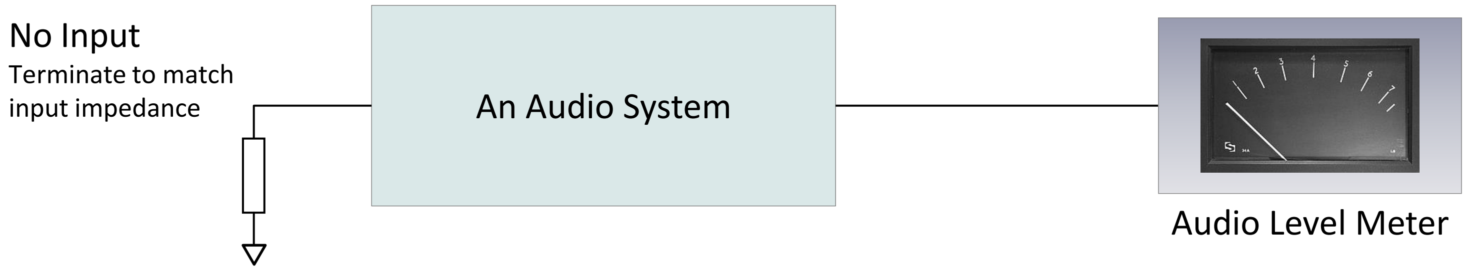
Noise Measurements – Analogue Video

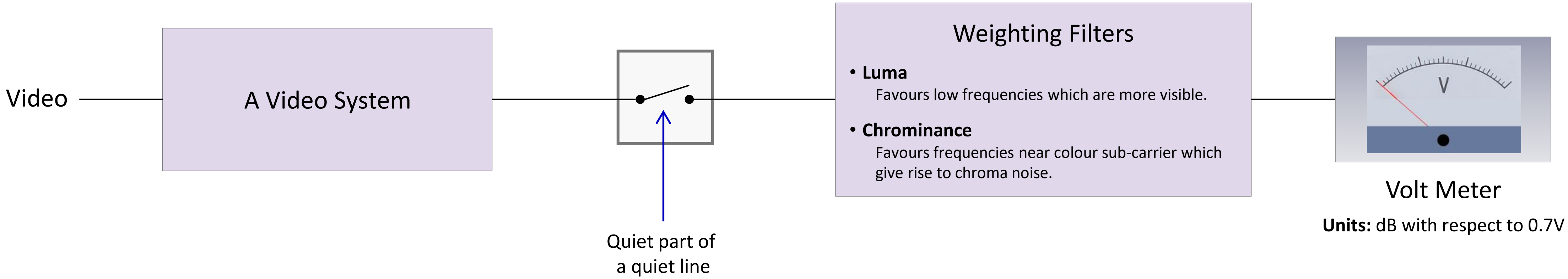
- There are usually 2 types of noise which are looked for:
  - **Random Noise** – usually measured above line frequency (15kHz)
  - **Coherent Noise** (such as mains hum) – Measured below (15kHz)
- Expressed in dB as the ratio between rms noise voltage and peak signal (0.7V).

$$20\log_{10}\left(\frac{V_{noise}(rms)}{0.7\ V}\right)$$

- To measure noise in the presence of video, a 'quiet' part of the video is gated out.
- Weighting filters can be applied:
  - **Luminance Weighting: bias** the reading towards **low frequency noise** which **looks worse**.
  - **Chrominance Weighting: bias** the reading toward **noise around colour subcarrier frequency** which gives rise to **low frequency colour noise**.

Noise Measurements – Audio vs Video

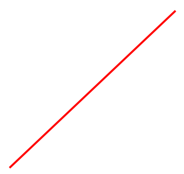




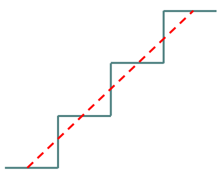
Noise Measurements for Digital Video

- The digitisation process add **quantisation noise** to the video.

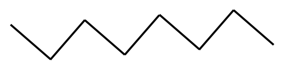
- This is the **difference** between the **analogue levels** and the **fixed levels** after the video has been **quantised** into digital levels.
  - 1024 available levels for a 10 bit system, 256 levels for an 8 bit system (although not all levels are used for the actual signal).



Analogue Video Signal



Quantised Video Signal



Difference – Quantising Noise

- Quantisation noise can be measured using a **ramp waveform** passed through a **digital system**.

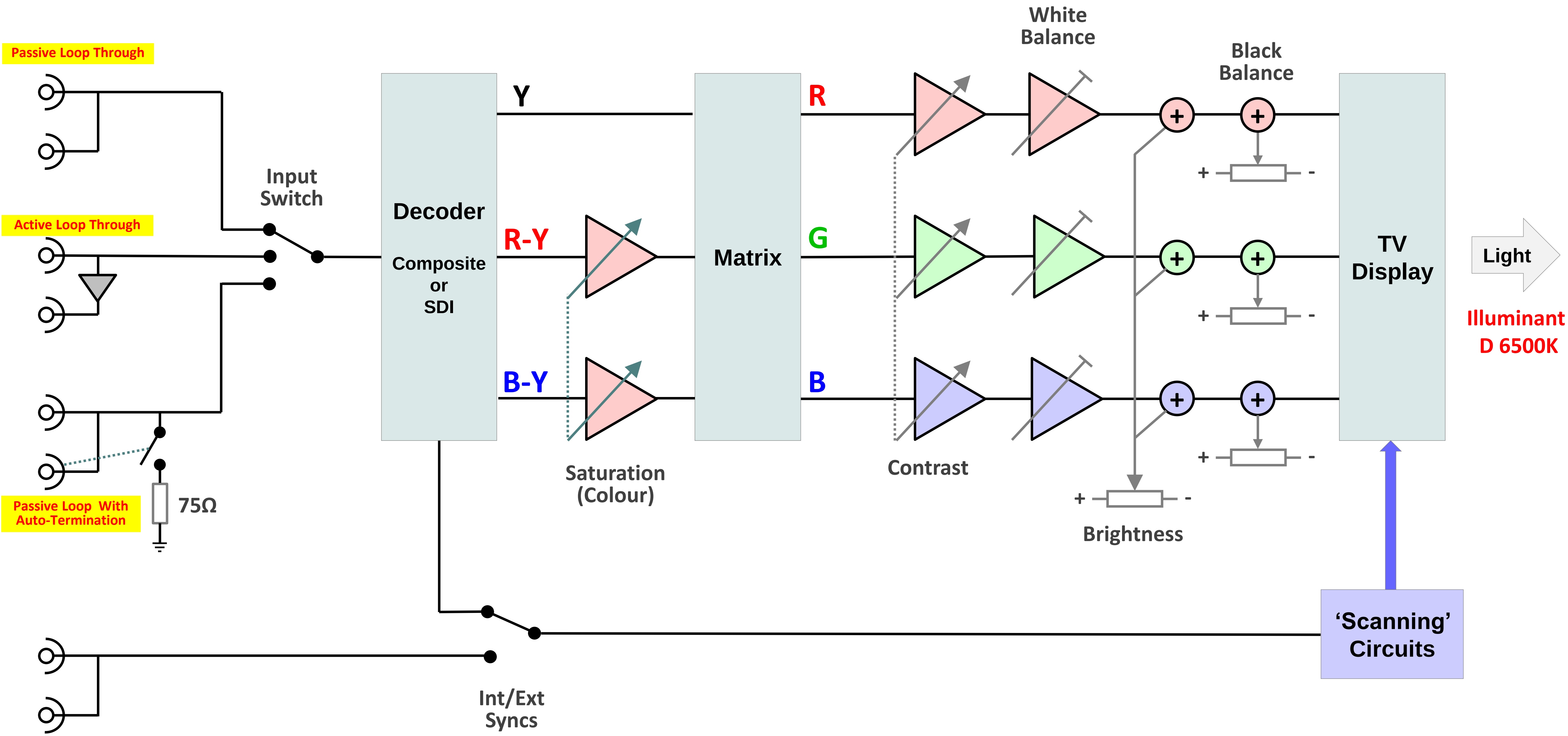
- The ramp is **nulled out** in the **test set** by adding a **perfect inverse analogue ramp**.

- The resulting **quantising noise** is measured using similar techniques to the those used for random noise.

Lecture Outline

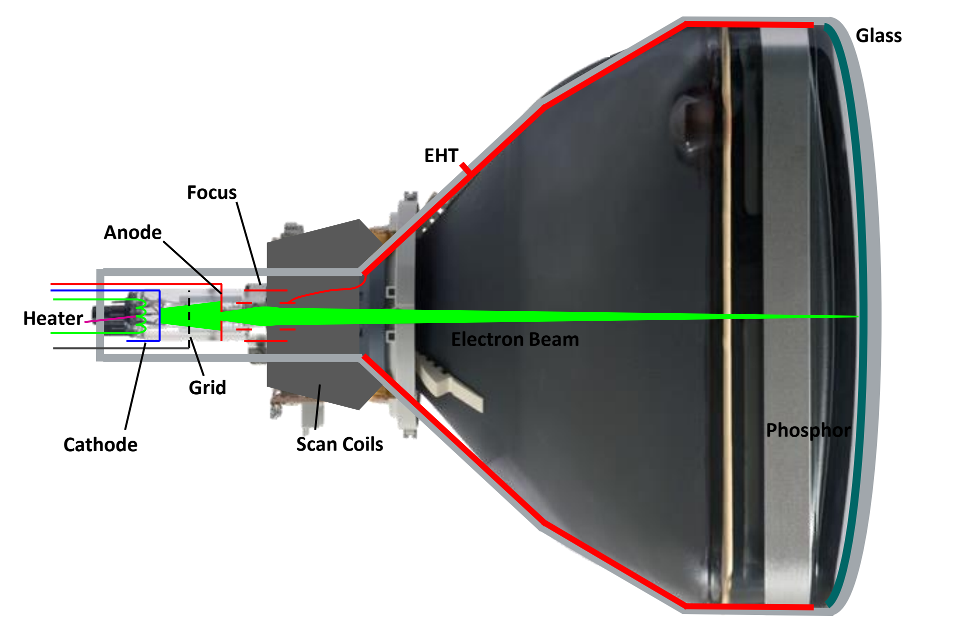
- Introduction (30 mins)
  - Why do we test? Tolerances
  - Picture quality assessment
  - Types of testing & Monitoring
  - Accuracy and calibration
  - Standards and specifications
  - Broadcast chain & measurement domains
  - Test equipment
- Signal Domain Tests (50 mins)
  - What can go wrong?
  - Analogue Video
  - Digital Video
- Video Domain Tests (50 mins)
  - What can go wrong?
  - Single channel tests
  - Component tests
  - Test waveforms and displays
  - Noise measurements
- Picture Monitor Grading & Setup (40 mins)
  - Revision of Monitor Principles
  - Monitor Grading
  - Monitor Setup – Adjustments and test signals

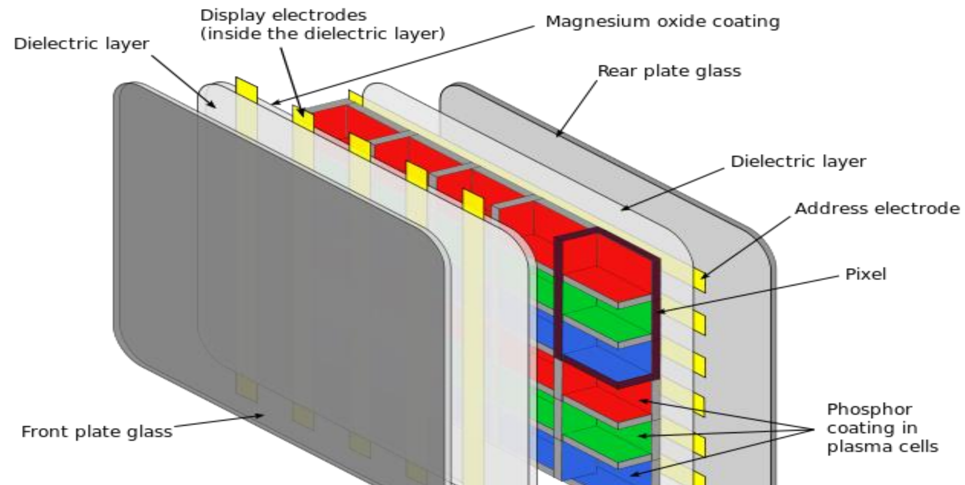
TV Monitor Block Diagram



TV Displays - Obsolete

- CRT
  - A glass structure containing a near **vacuum**.
  - An **electron gun** fires **a beam of electrons** at a **phosphor** on the rear of the screen.
  - The exited phosphor **emits light proportional to beam current**.
  - **Magnetic fields** from **2 sets of coils** are used to **scan** the beam horizontally and vertically.
- Plasma
  - Pixels consist of **phosphor coated glass capsules**, filled with **low pressure gas** (neon or xenon) which is excited by nearby electrodes.
  - The gas emits **ultra violet light** which **excites the phosphor**.
  - Brightness variations are created by **switching** the pixel on and off quickly (**P**ulse **W**idth **M**odulation).





A schematic matrix electrode configuration in an AC PDP

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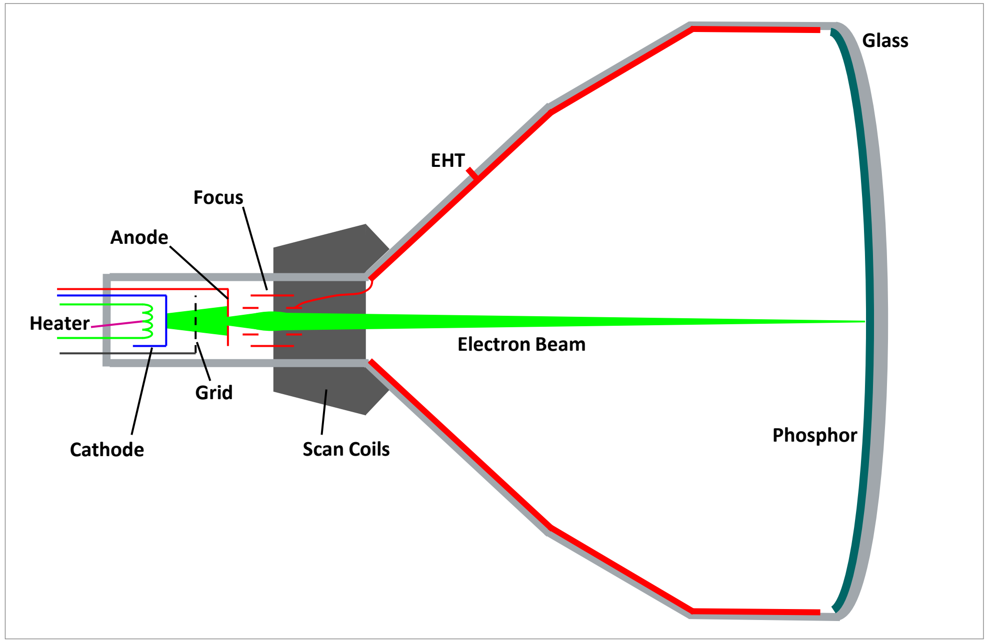
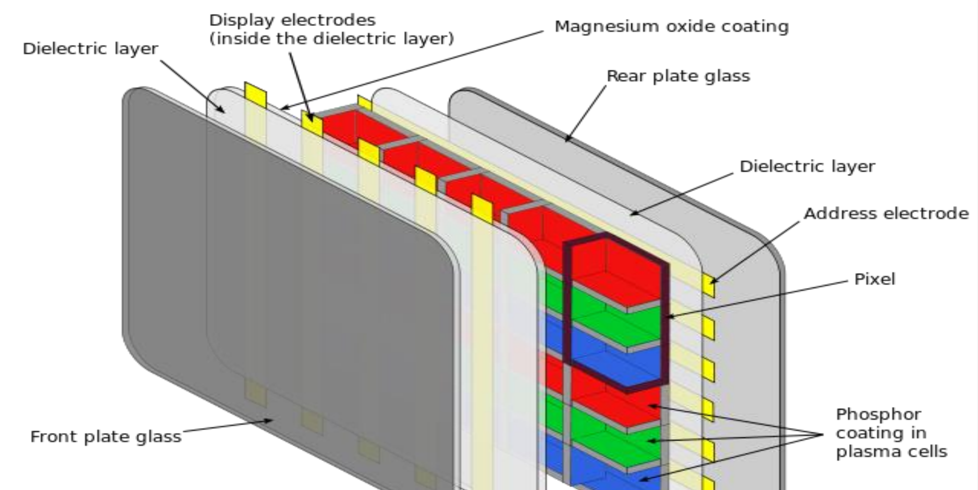
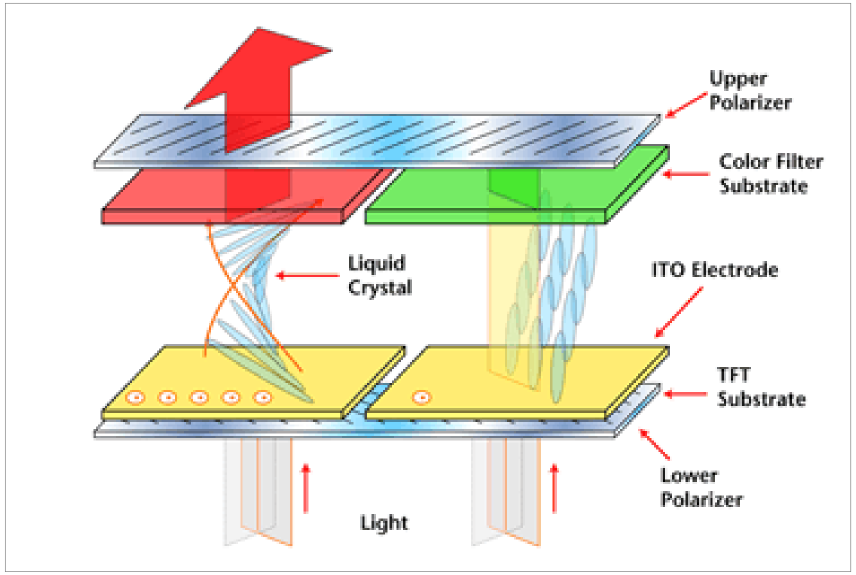


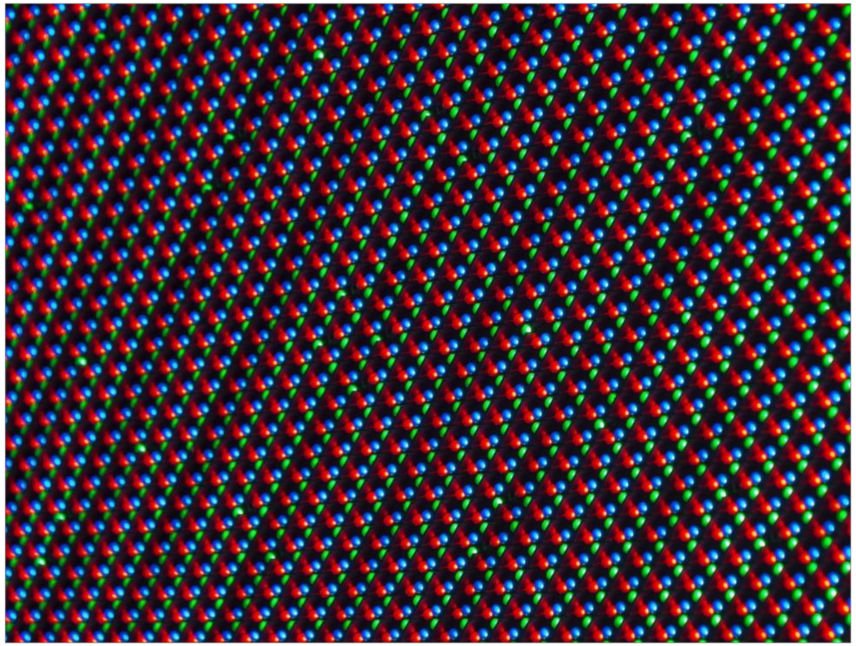


A schematic matrix electrode configuration in an AC PDP

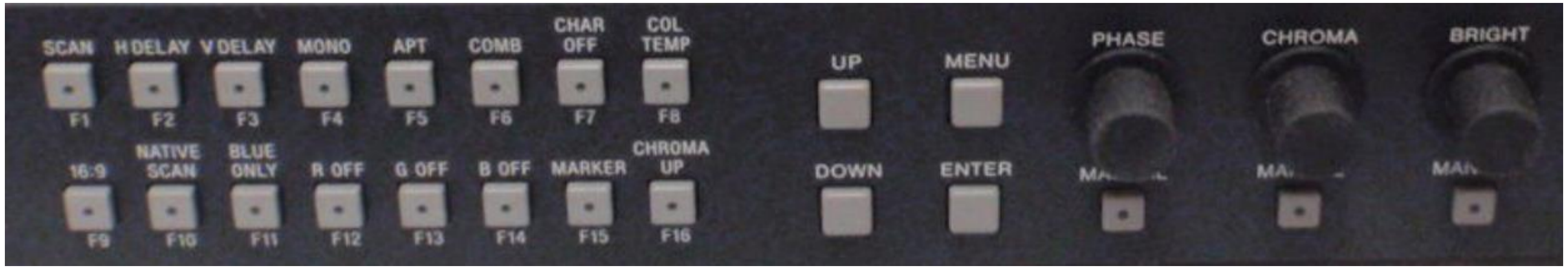
TV Displays

- LCD
  - Each pixel consists of **liquid crystal** between **2 polarizing filters** which are perpendicularly polarized.
  - There is a **backlight** behind the pixels – **cold-cathode fluorescent** or **LED** (These are sometimes called LED TVs for marketing purposes).
  - Without the liquid crystal, the backlight would be **blocked** by the filters.
  - The liquid crystal **twists the polarization** of light by **90°** allowing light to pass through.
  - An **electric field** from nearby electrodes changes the polarization of the liquid crystal. In the extreme case the polarization will be zero and the filters will block all the light. Varying the voltage will proportionally vary the light.
- LED
  - Each pixel is an Light Emitting Diode.
  - Very large display screens can be made with arrays of individual LEDs (Light Emitting Diodes).
  - Or they can be made using integrated techniques.
  - The main technologies are OLED (Organic LED) and MicroLED.





Monitor Facilities



- Professional and broadcast monitors have extra facilities compared to a domestic TV. For example:
  - External Syncs
  - Under-scan
  - Adjustable white and black balance
  - Adjustable and preset brightness, contrast and saturation
  - Monochrome button
  - Switching on/off individual R, G, B channels
  - Blue only button (or blue to monochrome)
  - Loop-Thru Inputs
  - Display shift (pulse cross)
    - Allows you to see the blanking interval

Monitor Grading (See EBU Tech 3320)

- Grade 1 – for high-grade technical quality evaluation:
  - For camera colour control, colour grading, lighting control etc.
  - Must possess at least the quality properties of the equipment to be assessed.
  - State-of-the-art technology - Artefacts should not be masked nor additional artefacts introduced.
  - A ‘measuring instrument’ for visual evaluation of image quality - Desirable: Native scanning mode of the signal - progressive or interlaced.
- Grade 2
  - For preview, control walls, edit suites, control rooms - if no picture quality manipulation done.
  - Similar facilities to a Grade 1 monitor.
  - Wider tolerances on its specification than a Grade 1 monitor, so cheaper, smaller, lighter.
- Grade 3
  - For observation/presence – audio production, dialogue dubbing, commentary positions, audience displays.
  - Similar to high end domestic/consumer displays – extra features: Inputs, mounting, robustness.

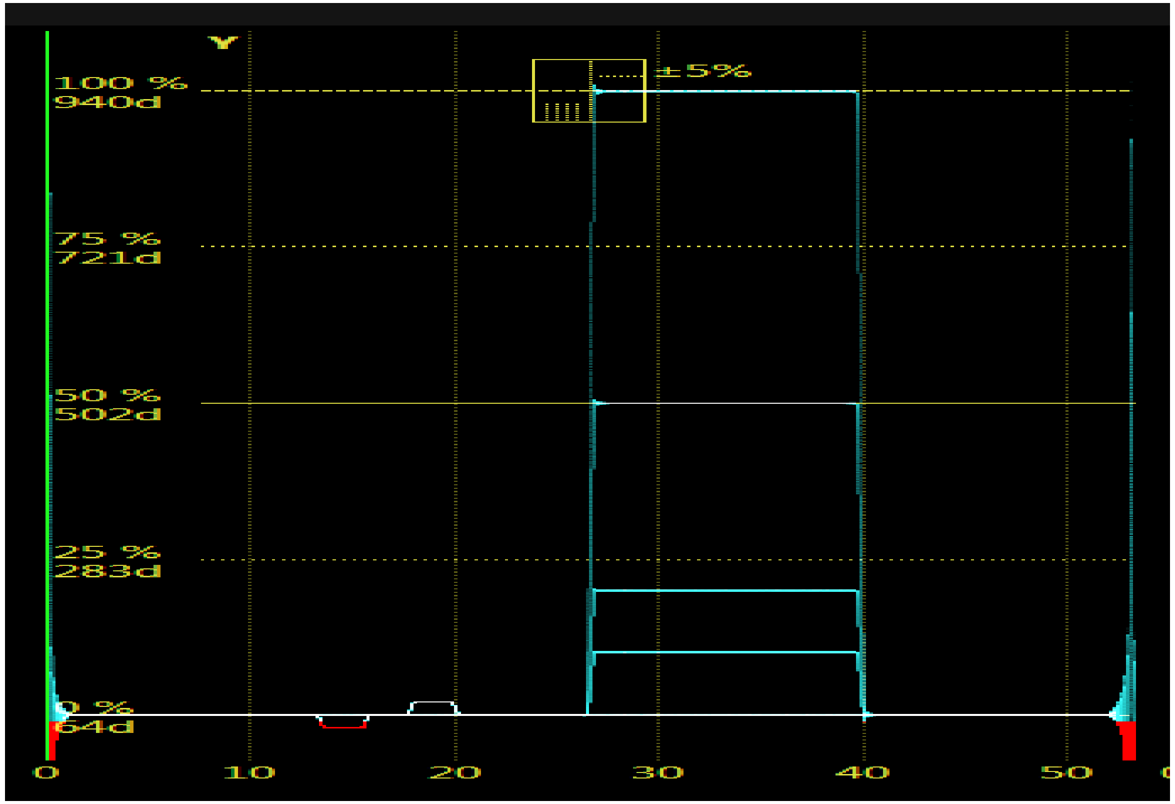
Monitor Adjustments - Overview

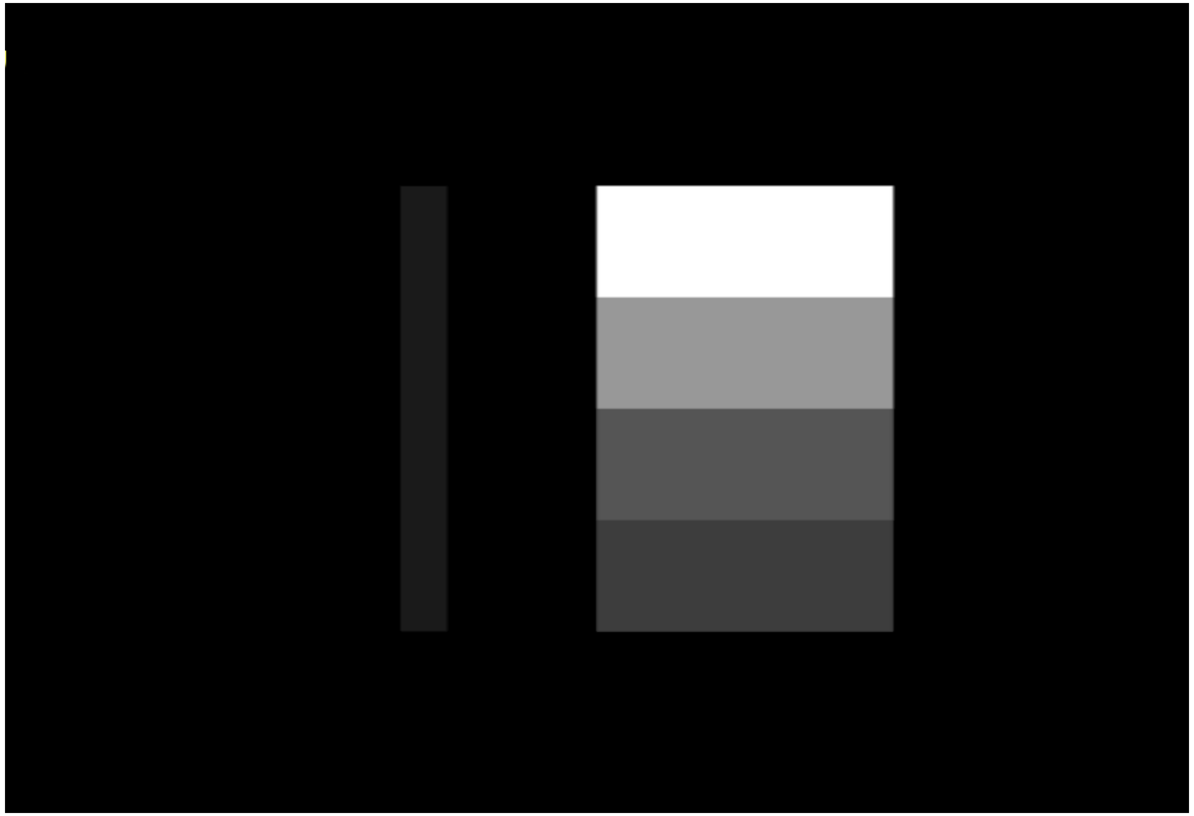
- Basic Adjustments
  - Brightness
  - Contrast
  - Saturation (Colour)

4 most commonly made adjustments

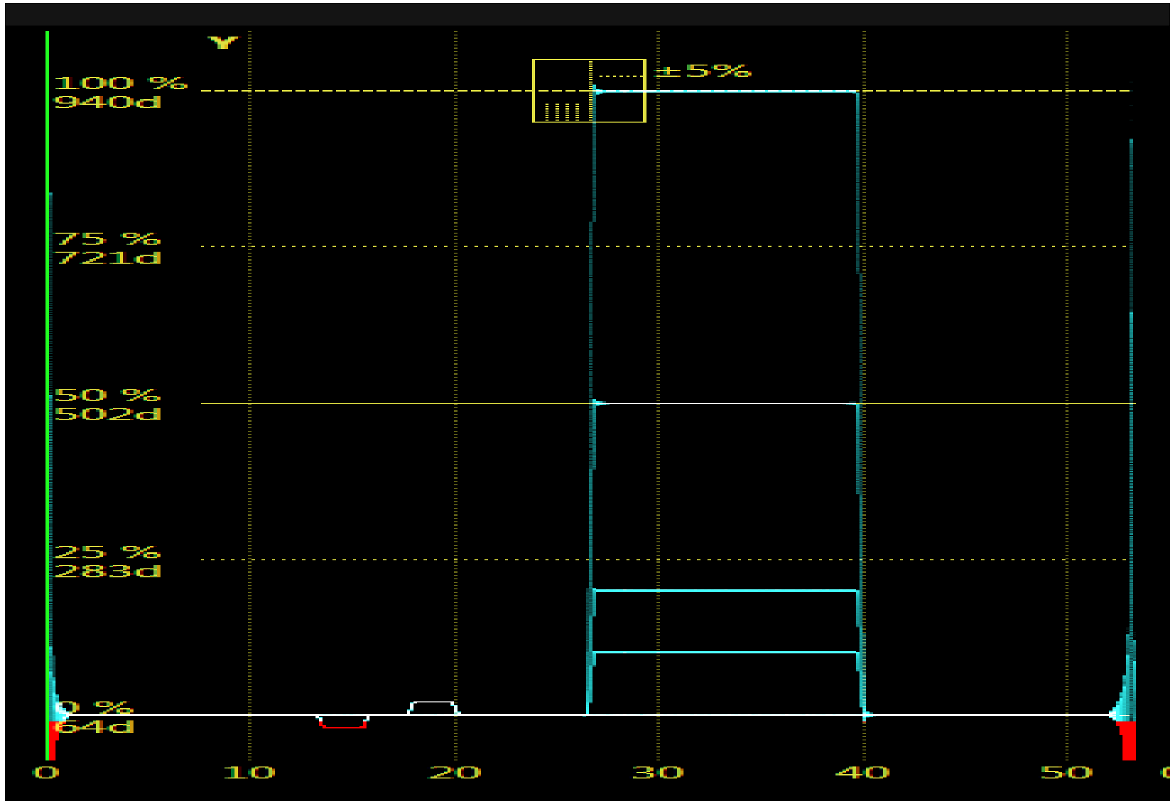
- Colour Balance
  - White balance - individual RGB contrast (also called gain)
  - Black balance - individual RGB brightness (also called D.C. level, lift/sit, bias)
- Other checks
  - Especially in Grade 3 monitors, make sure any aperture (sharpness) controls are not set too high (which results in overshoots/ringing on edges and over emphasizes noise).
  - Turn off unnecessary video processing (‘smooth motion’ etc.)
- CRT Adjustments (historical interest only)
  - Picture geometry (grill/crosshatch test patter), Picture size (test card), Purity (red field), Convergence – individual RGB geometry (grill/crosshatch test pattern)

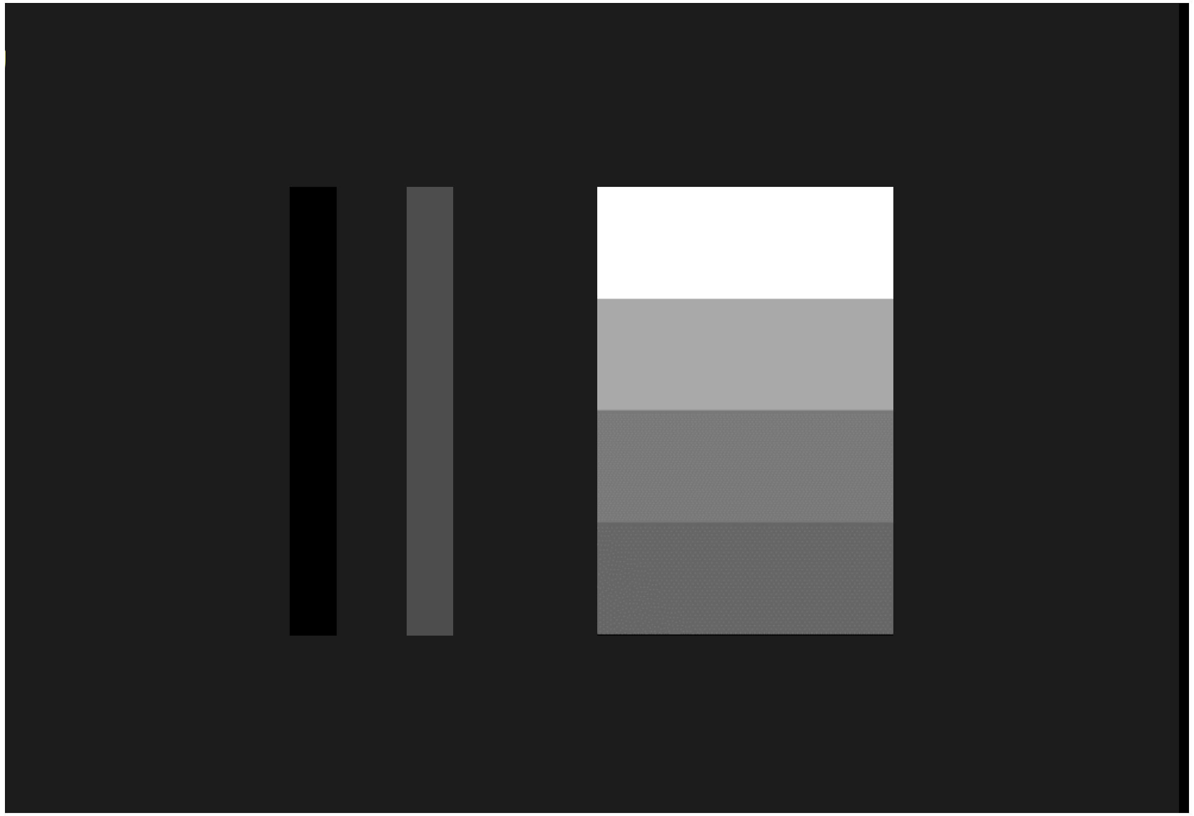
Monitor Test Signals - PLUGE





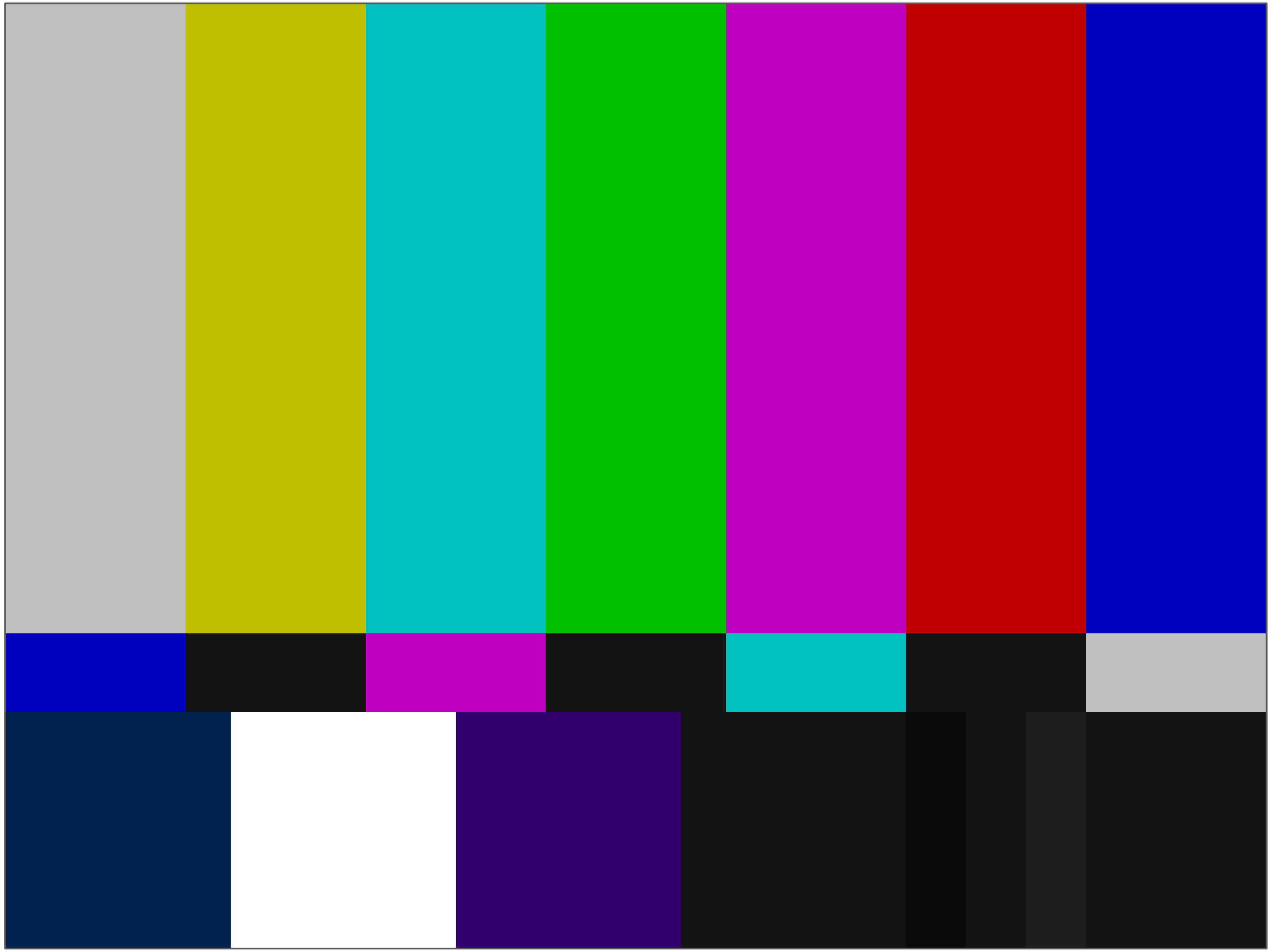
Monitor Test Signals - PLUGE



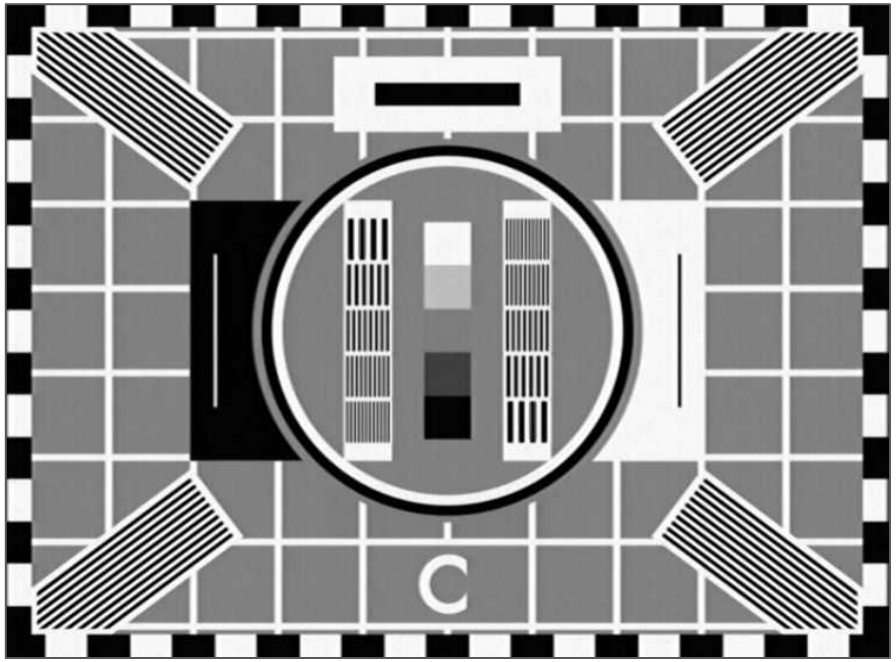


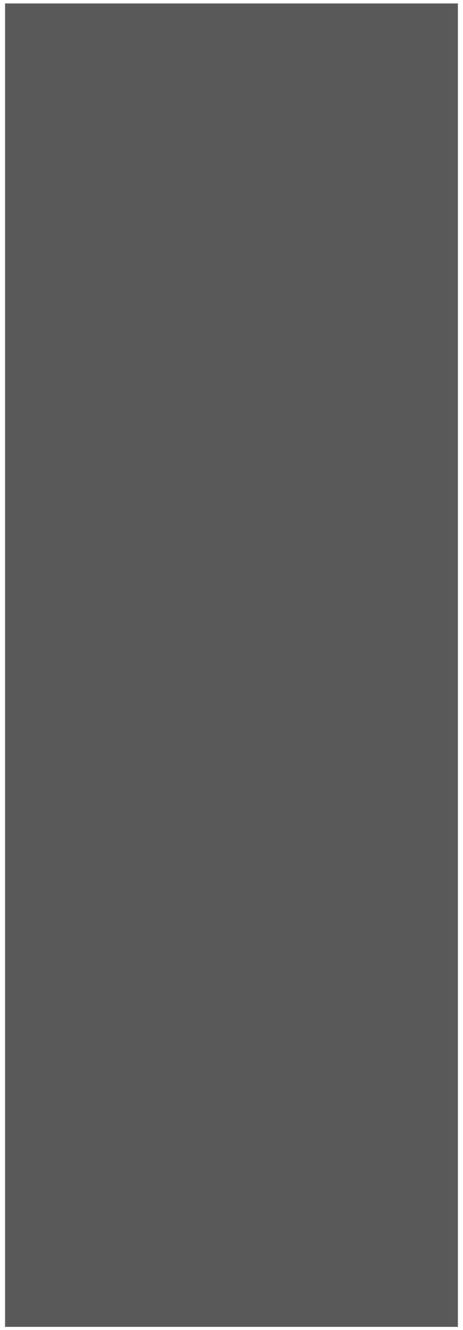
Monitor Test Signals - PLUGE in SMPTE Colour Bars

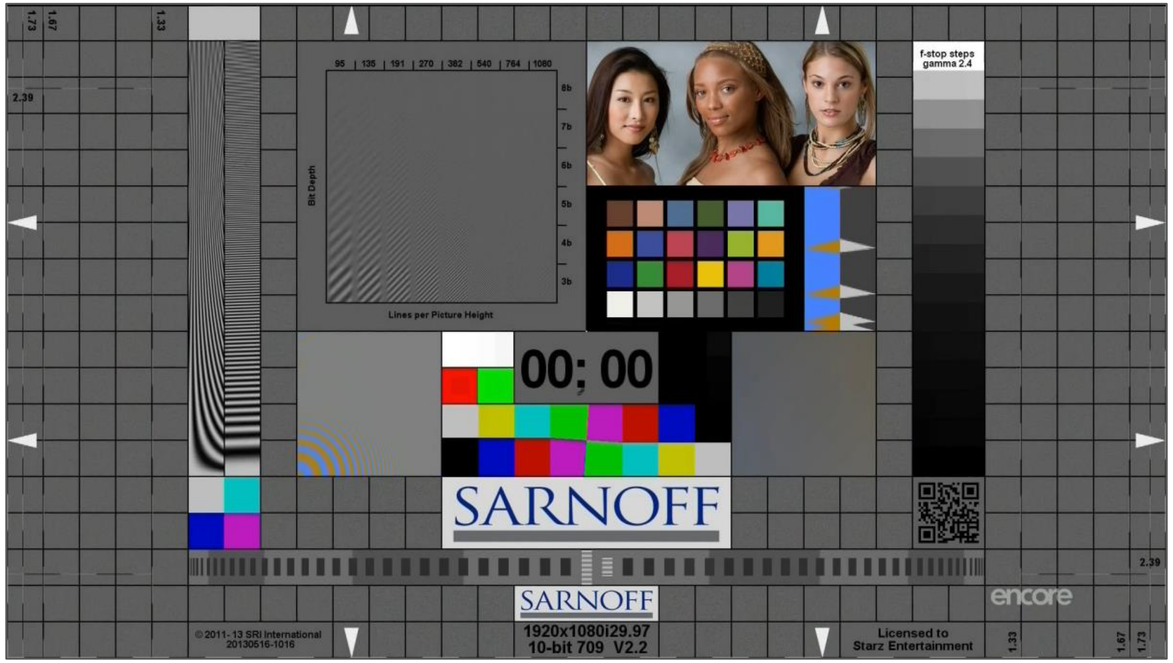
- SMPTE Colour Bars were designed for NTSC analogue colour systems.
- Often available from test signal generators in other formats.
- Usefully, they contain a PLUGE section.



Monitor Test Signals – Test Cards





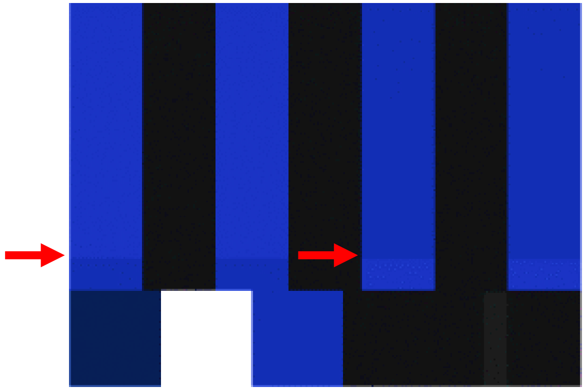


Basic Monitor Adjustments

- Brightness – DC Shift
  - PLUGE – adjust so +2% strip can be seen, -2% strip cannot.
- Contrast – Gain
  - PLUGE – make white parts acceptably bright without clipping (or match to a reference using a spot light meter or monitor analyser).
- Saturation
  - Use 100% Bars
  - Look at blue signal only. Adjust until blue and white bars look equally bright
  - Ideally bars with a white strip such as SMPTE bars or 'Tartan Bars' so that blue and white are adjacent.

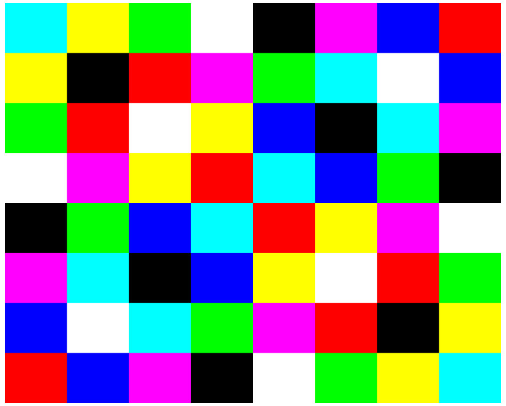
Brightness Correct

Brightness too High



Saturation too low

Saturation correct

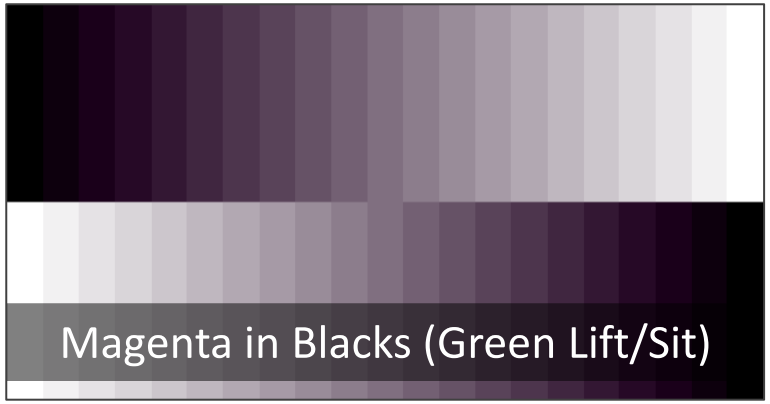


Colour Balance / Greyscale

- Colour Balance used to done manually:
  - Adjust Brightness and Contrast using PLUGE.
  - For Colour Balance use PLUGE or Greyscale.
  - Compare monitor to be adjusted to a monochrome Illuminant. D65 reference monitor.
  - Adjust white and black balance controls for a visual match.

- Use a Colour Analyser:
  - Use suitable test signal. Usually generated by the analyser.
  - Connect analyser probe to monitor screen.
  - Follow analyser instructions and adjust white and black balance.

- Use a Monitor Probe:
  - Connect monitor probe to monitor screen.
  - Find the right menu and press the 'auto adjust button'.



Monitor Analyzer

- Can be used to adjust a broadcast monitor.
- Can be used to adjust a computer display usually via software/driver.

References

ITU-R BT.500-13	Methodology for the subjective assessment of the quality of television pictures
ITU-R BT.814	Specifications and alignment procedures for setting of brightness and contrast of displays
ITU-R BT.815	Specification of a signal for measurement of the contrast ratio of displays
ITU-R BT.2022	General viewing conditions for subjective assessment of quality of SDTV and HDTV television pictures on flat panel displays.
EBU Tech 3320	User Requirements for Video Monitors in Television Production
EBU Tech 3263	Specification of Grade 1 Colour Production Monitors (CRT)
EBU Tech 3325	Methods for the Measurement of the Performance of Studio Monitors

The End